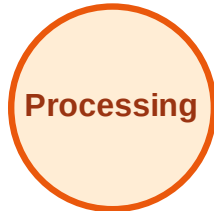


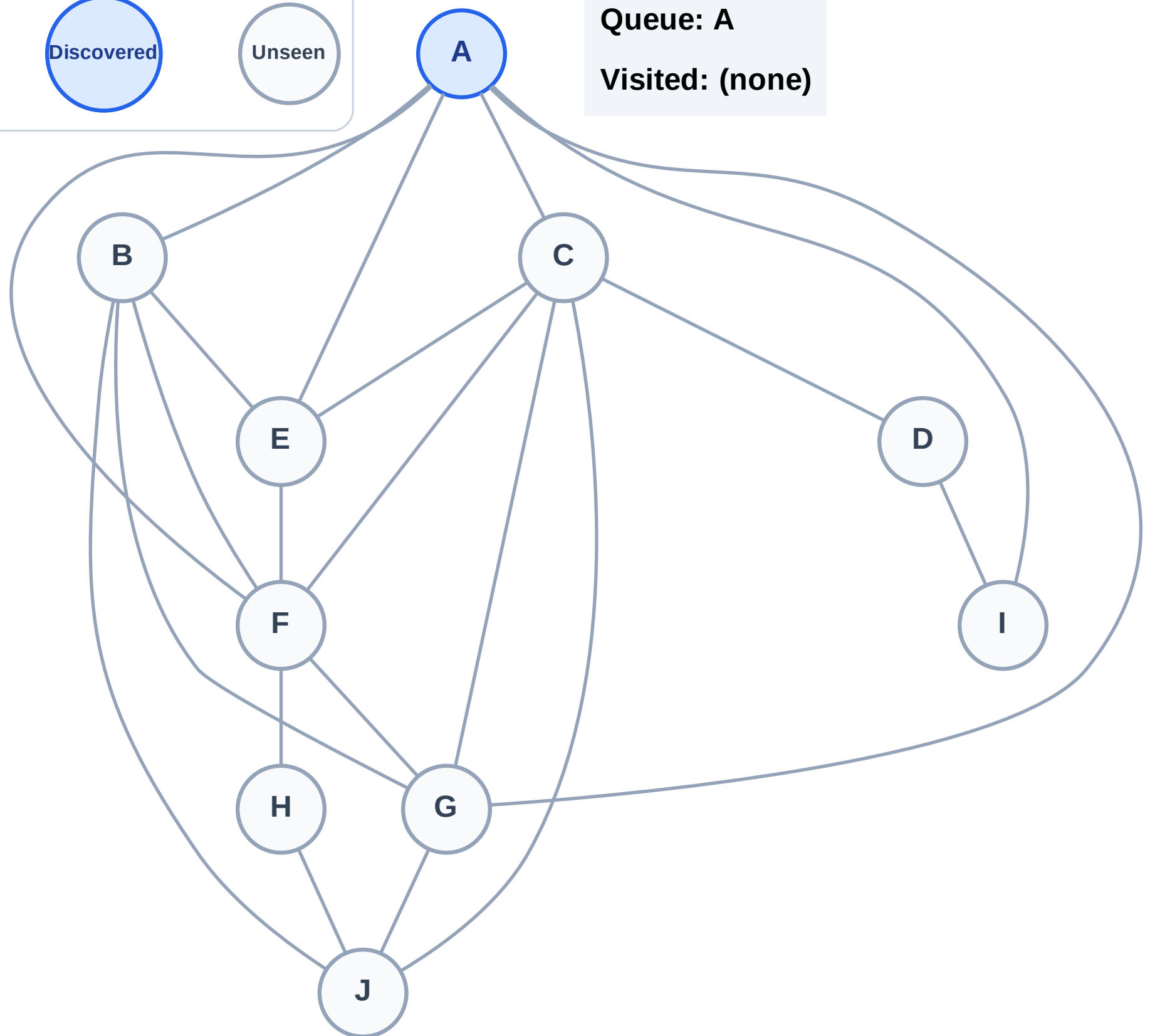
BFS Traversal

Step 1: Start at A

Legend



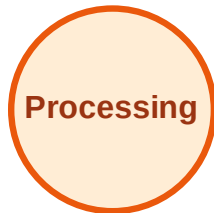
Queue: A
Visited: (none)



BFS Traversal

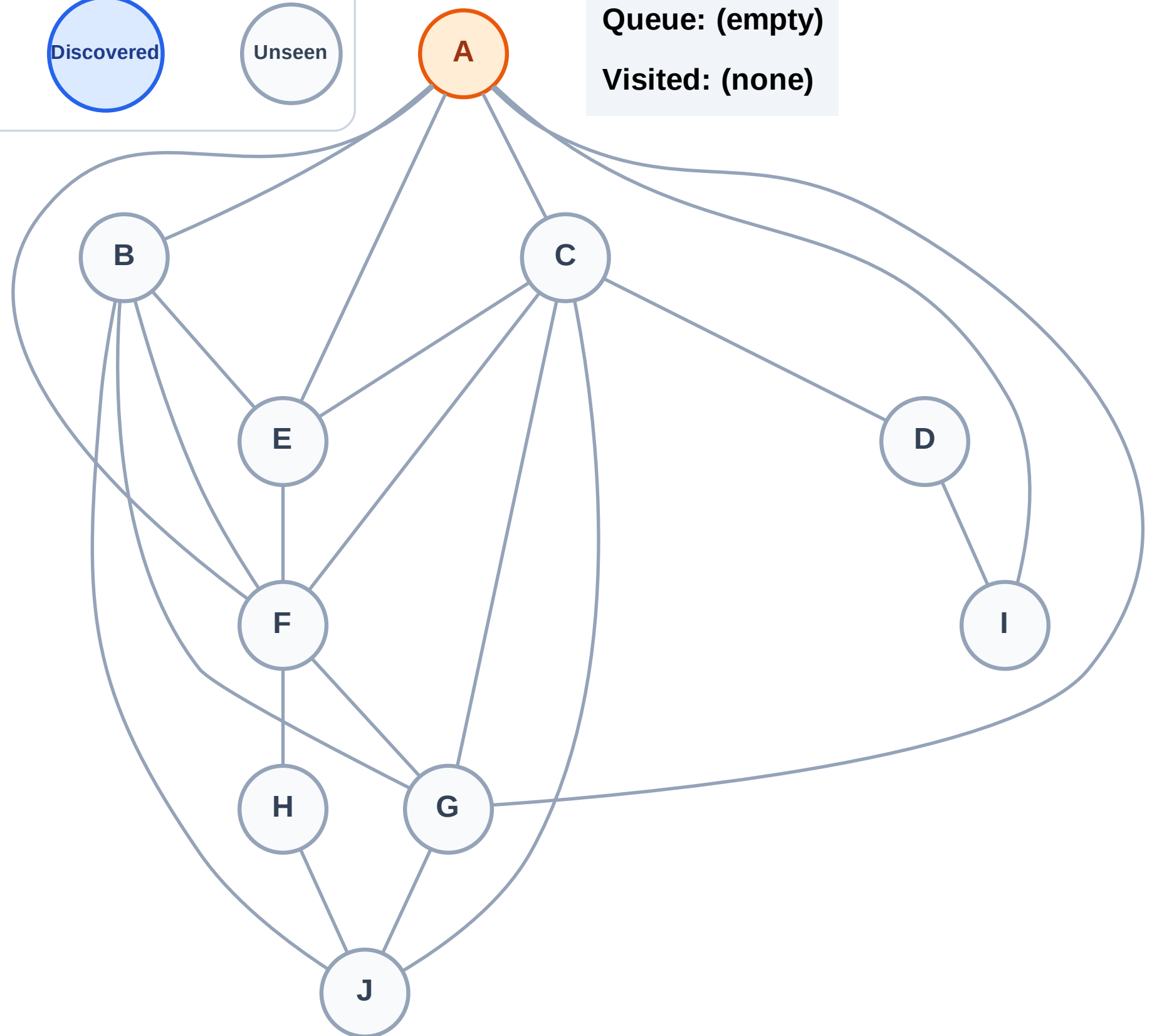
Step 2: Process node A

Legend



Queue: (empty)

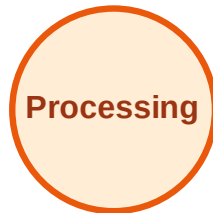
Visited: (none)



BFS Traversal

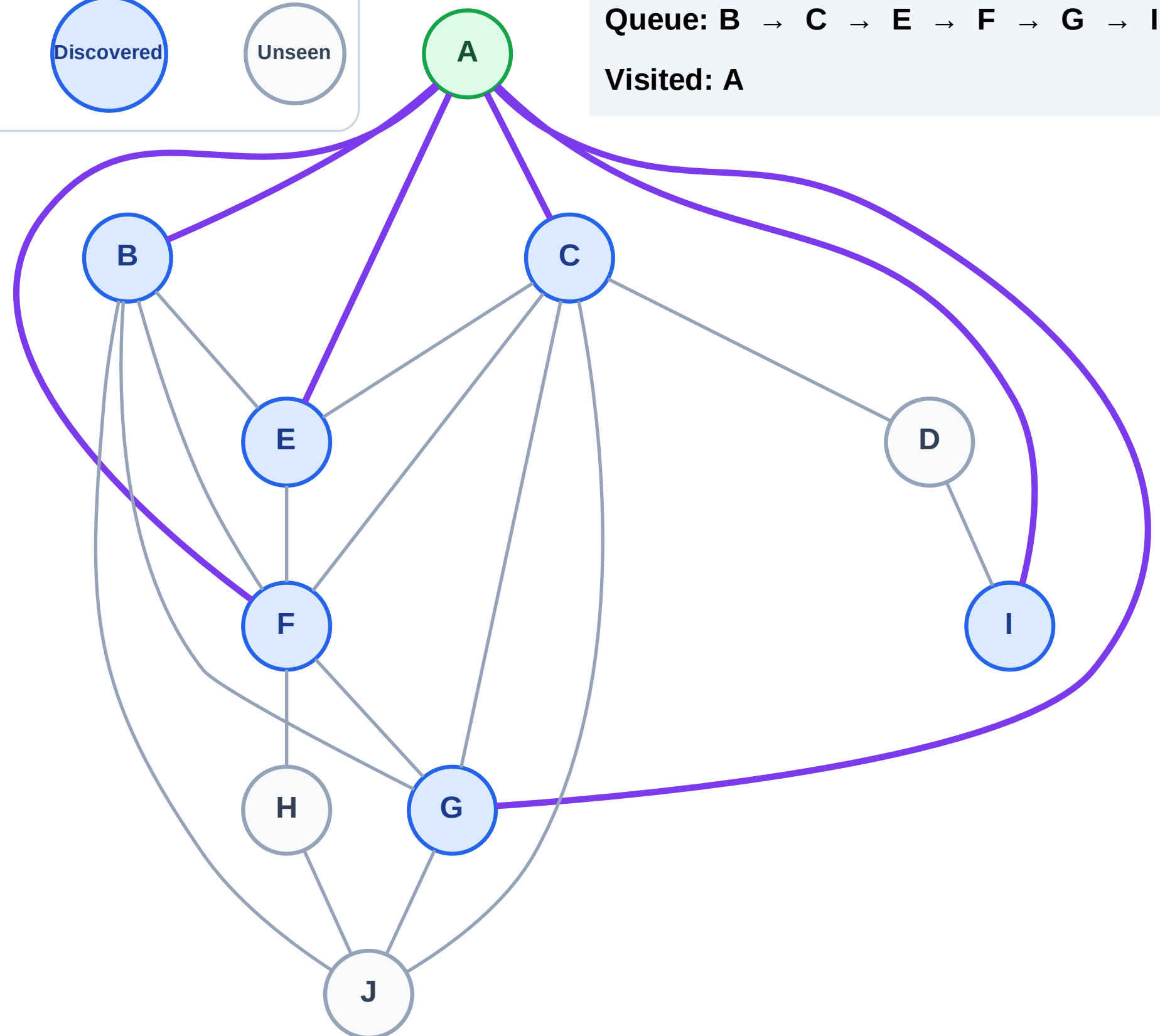
Step 3: Discover B, C, E, F, G, I from A

Legend



Queue: B → C → E → F → G → I

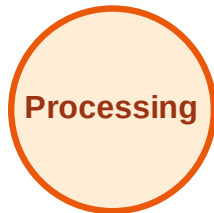
Visited: A



BFS Traversal

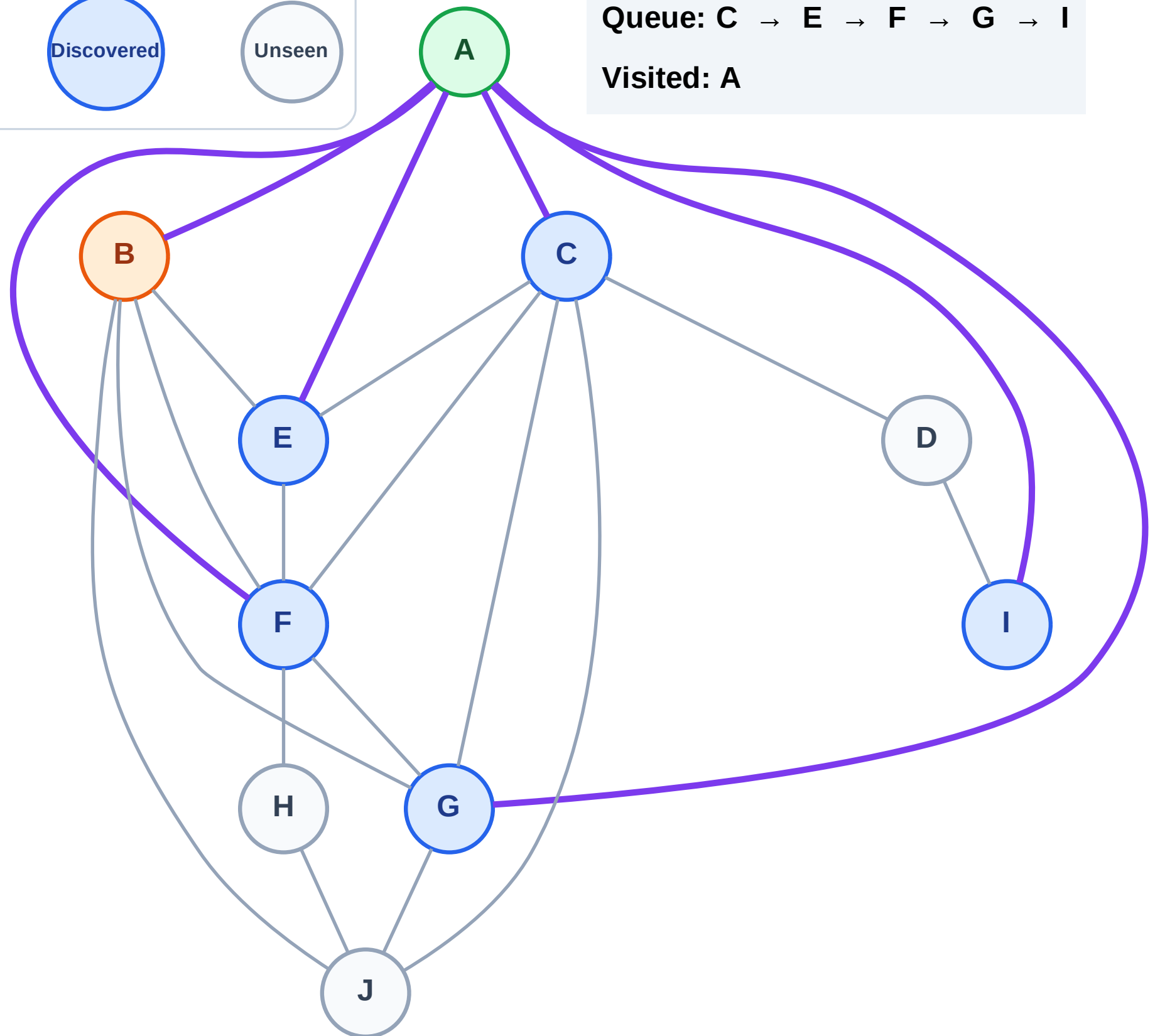
Step 4: Process node B

Legend



Queue: C → E → F → G → I

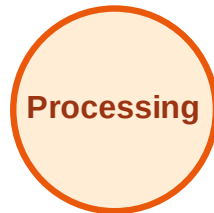
Visited: A



BFS Traversal

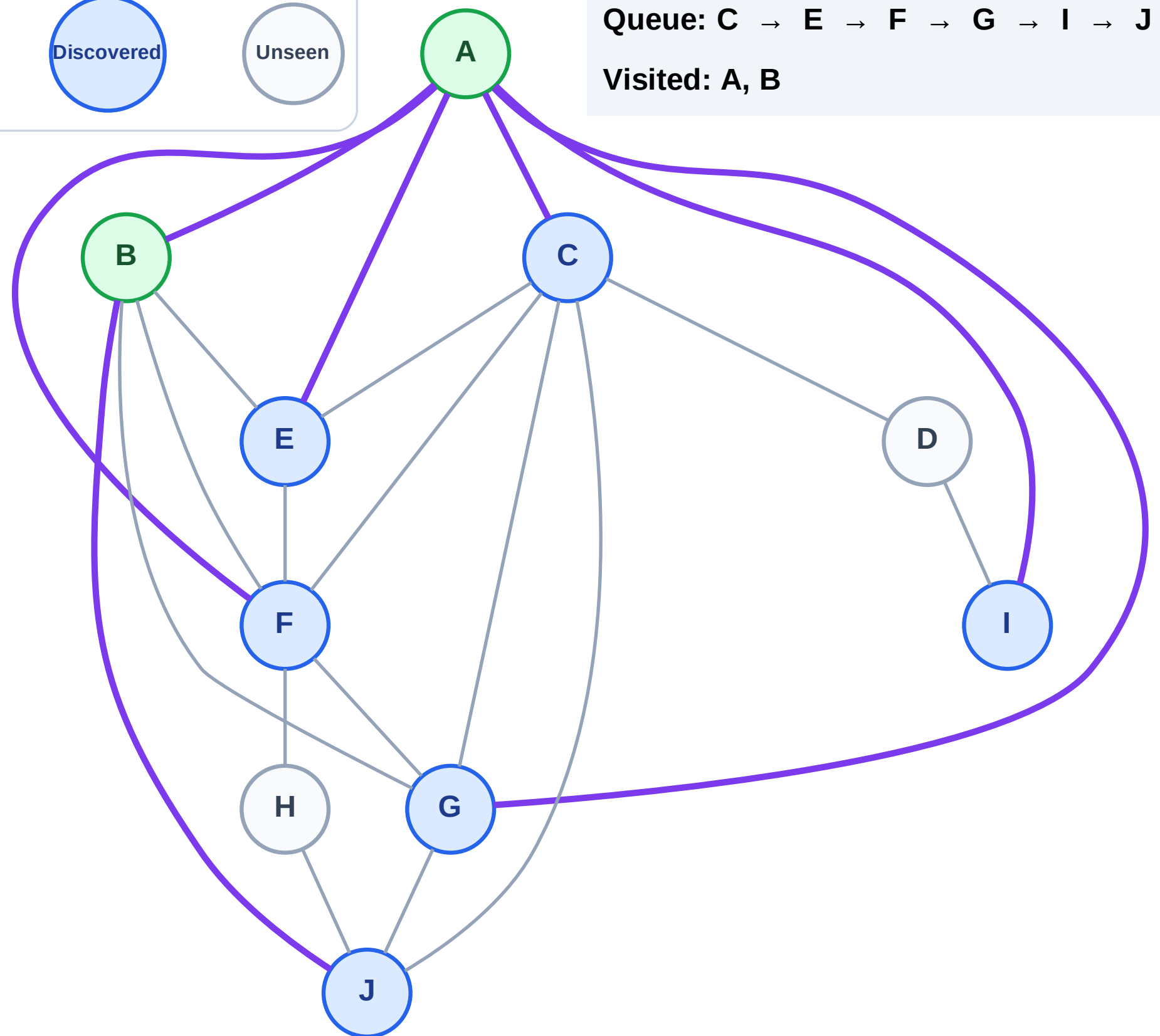
Step 5: Discover J from B

Legend



Queue: C → E → F → G → I → J

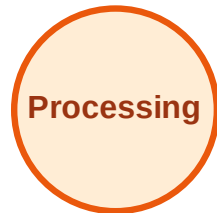
Visited: A, B



BFS Traversal

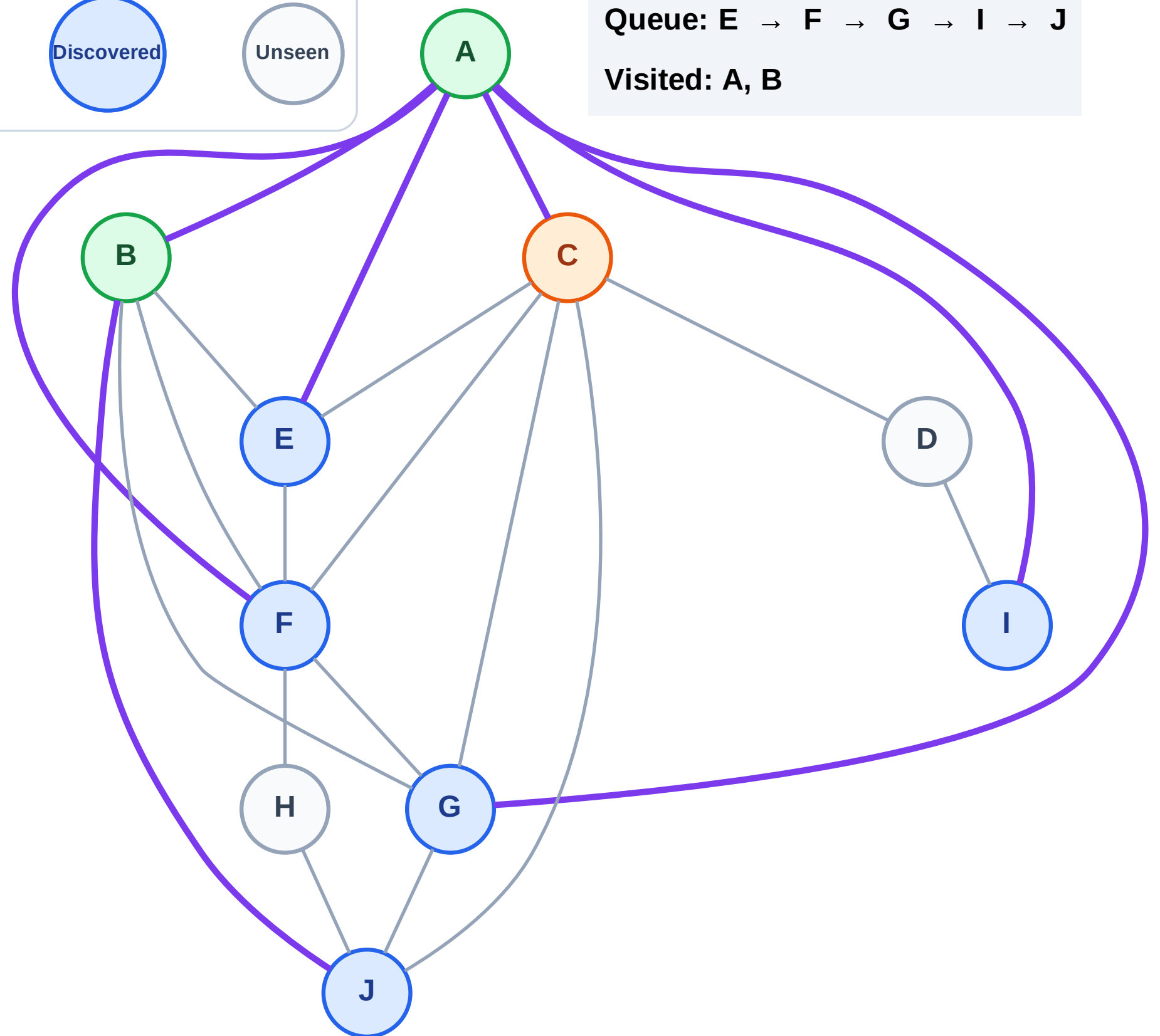
Step 6: Process node C

Legend



Queue: E → F → G → I → J

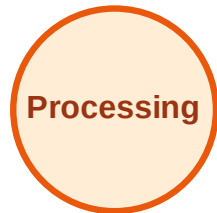
Visited: A, B



BFS Traversal

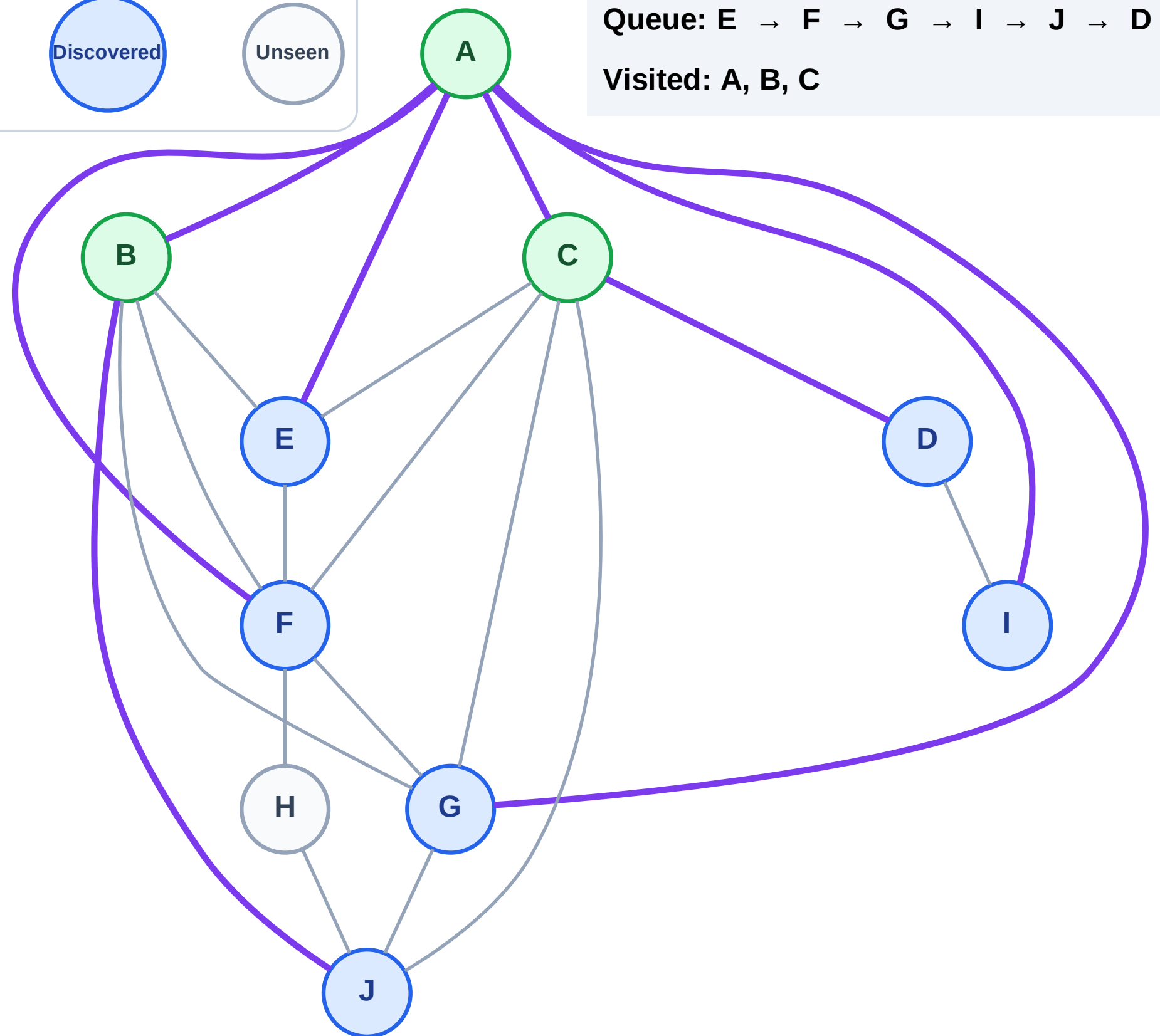
Step 7: Discover D from C

Legend



Queue: E → F → G → I → J → D

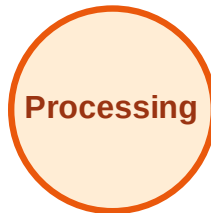
Visited: A, B, C



BFS Traversal

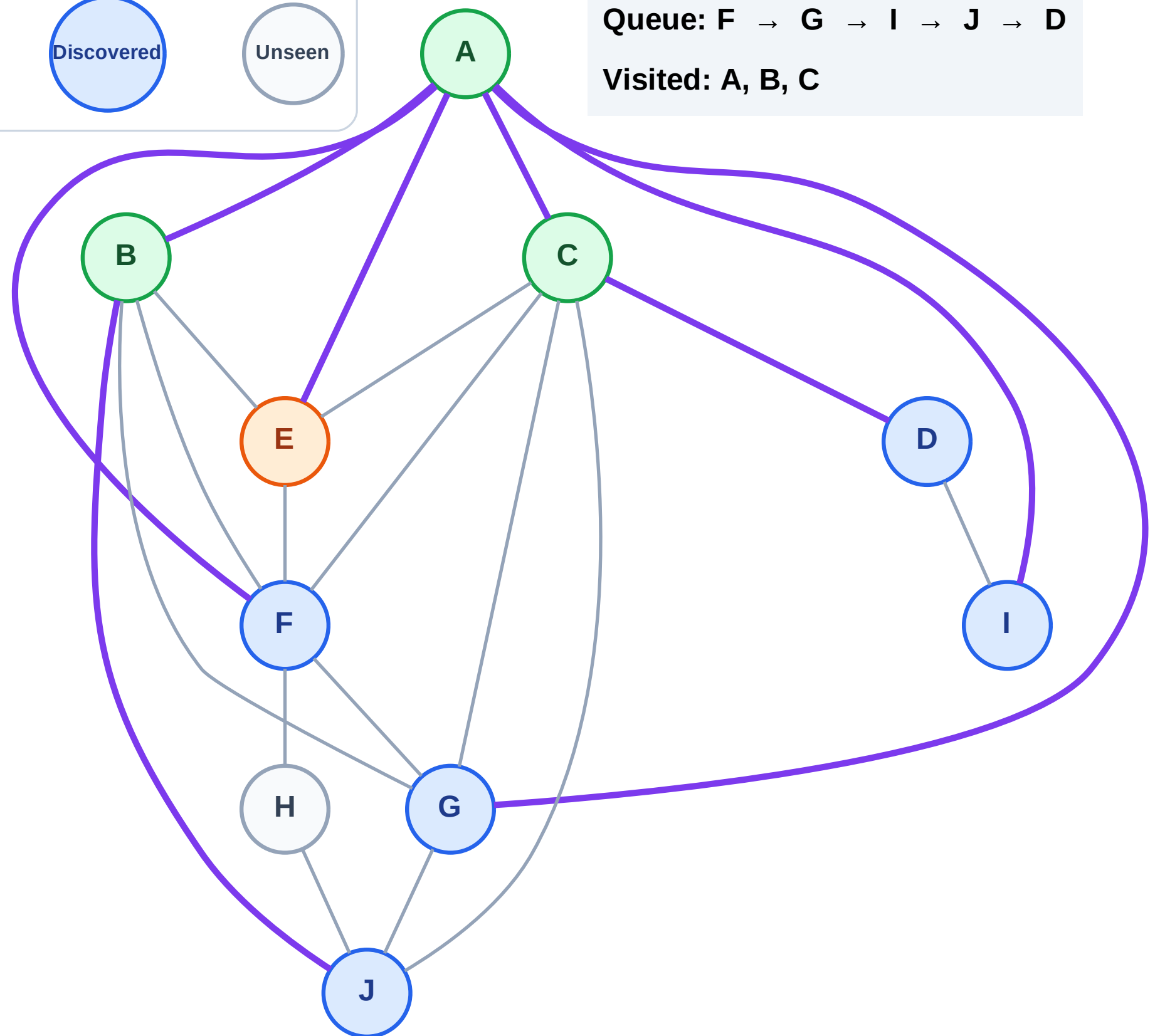
Step 8: Process node E

Legend



Queue: F → G → I → J → D

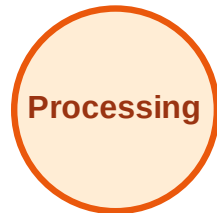
Visited: A, B, C



BFS Traversal

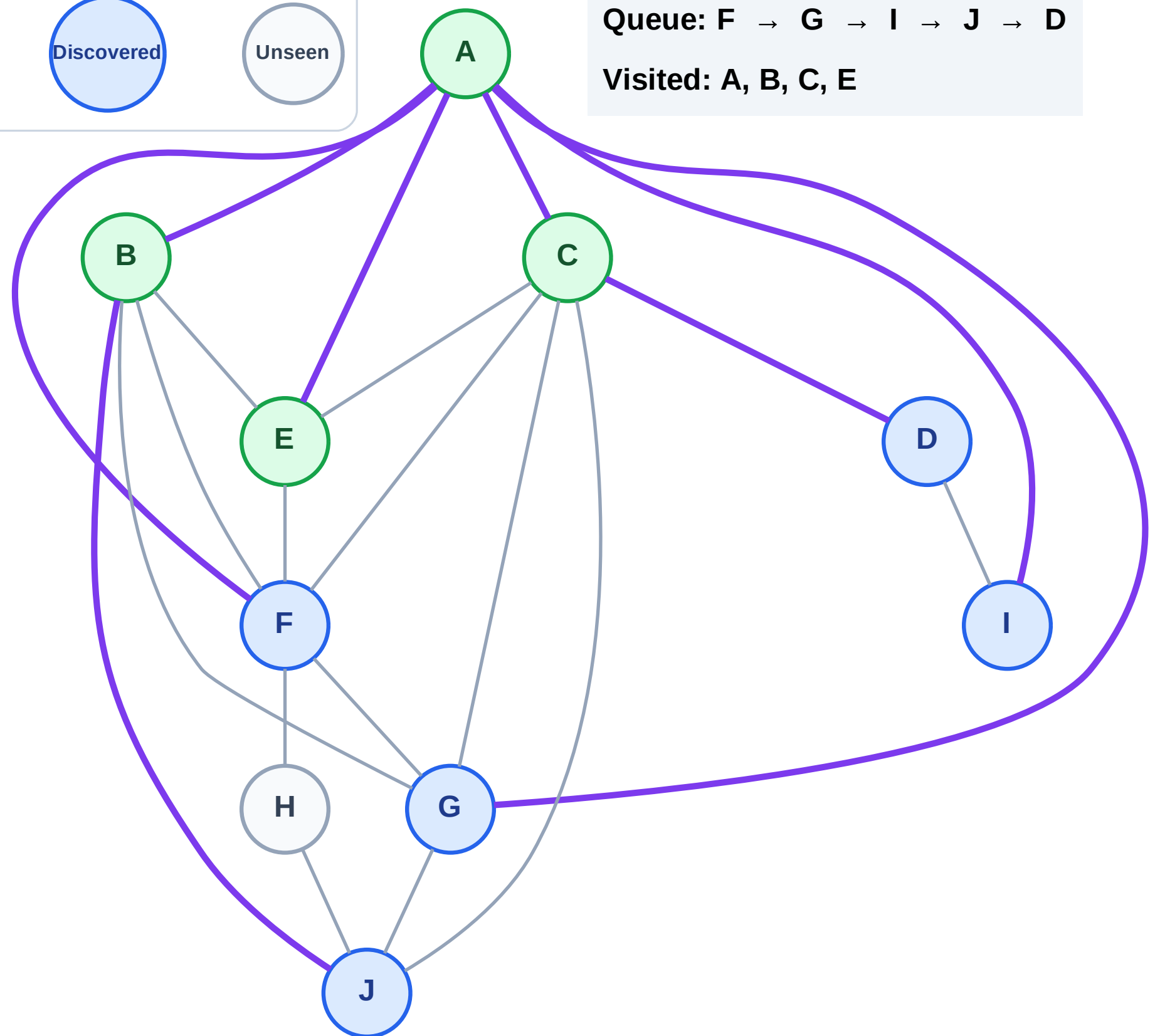
Step 9: No new neighbors from E

Legend



Queue: F → G → I → J → D

Visited: A, B, C, E



BFS Traversal

Step 10: Process node F

Legend

Complete

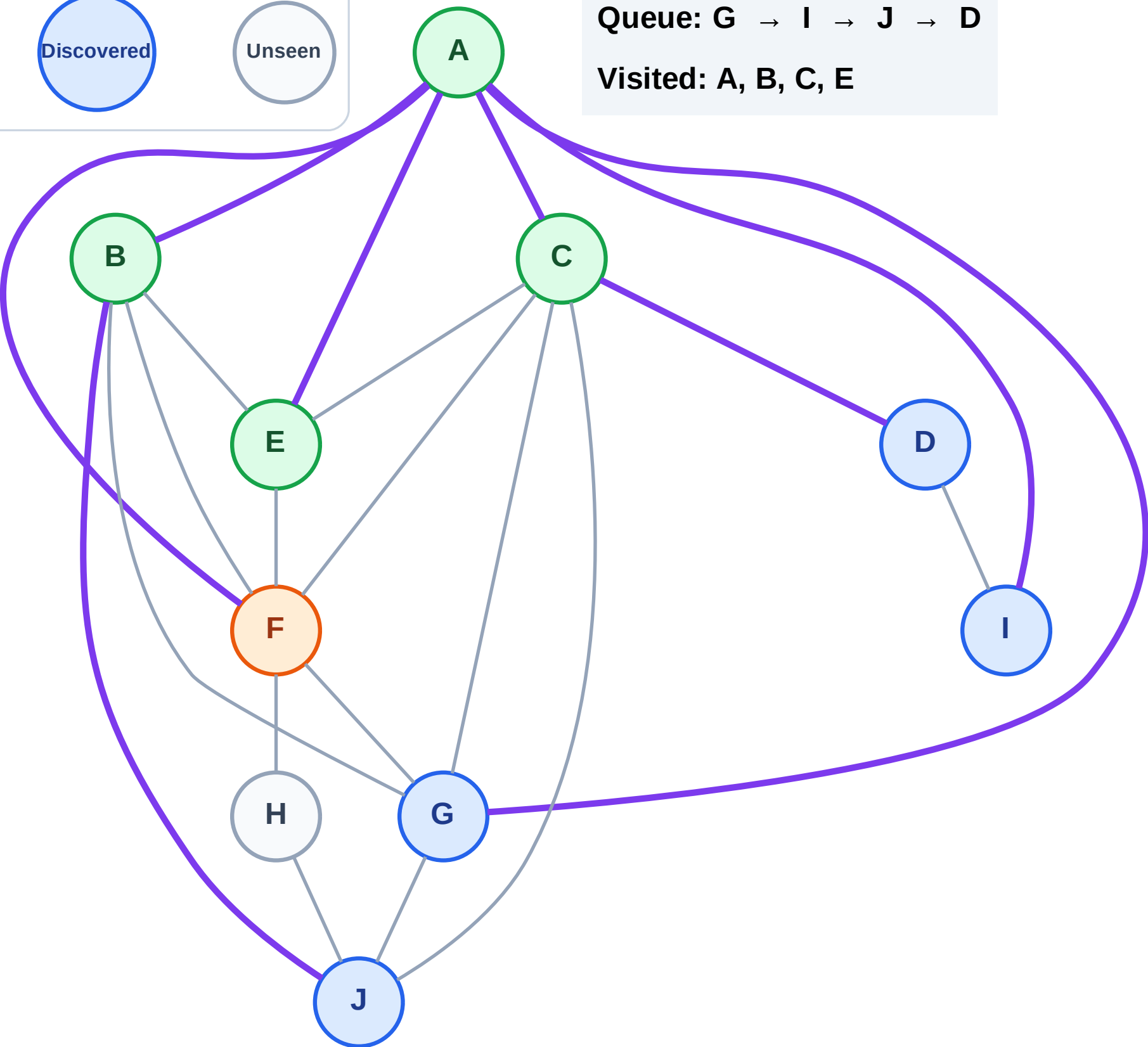
Processing

Discovered

Unseen

Queue: G → I → J → D

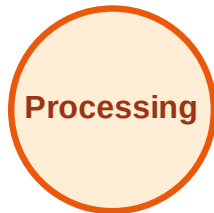
Visited: A, B, C, E



BFS Traversal

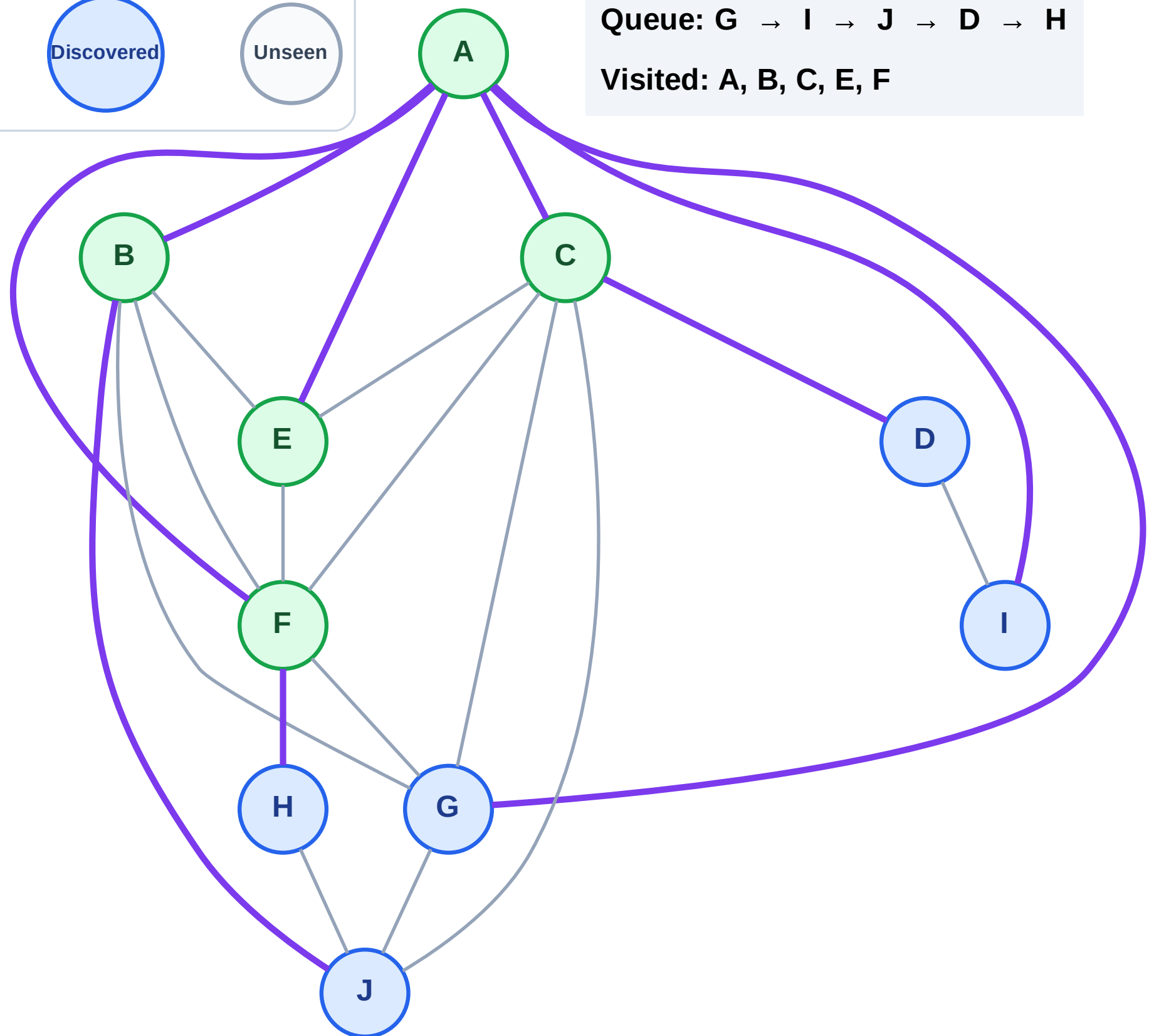
Step 11: Discover H from F

Legend



Queue: G → I → J → D → H

Visited: A, B, C, E, F



BFS Traversal

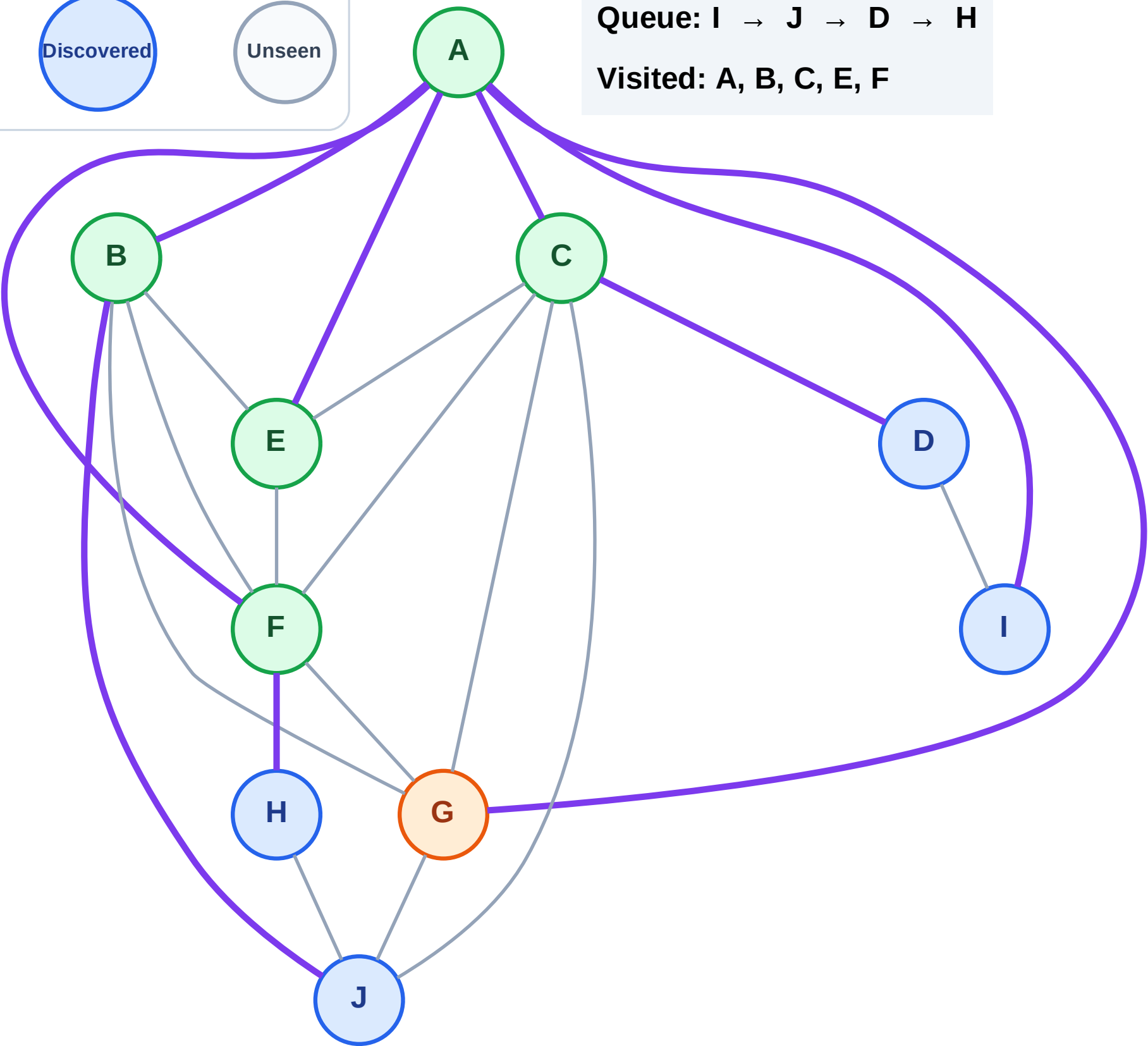
Step 12: Process node G

Legend



Queue: I → J → D → H

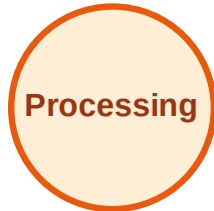
Visited: A, B, C, E, F



BFS Traversal

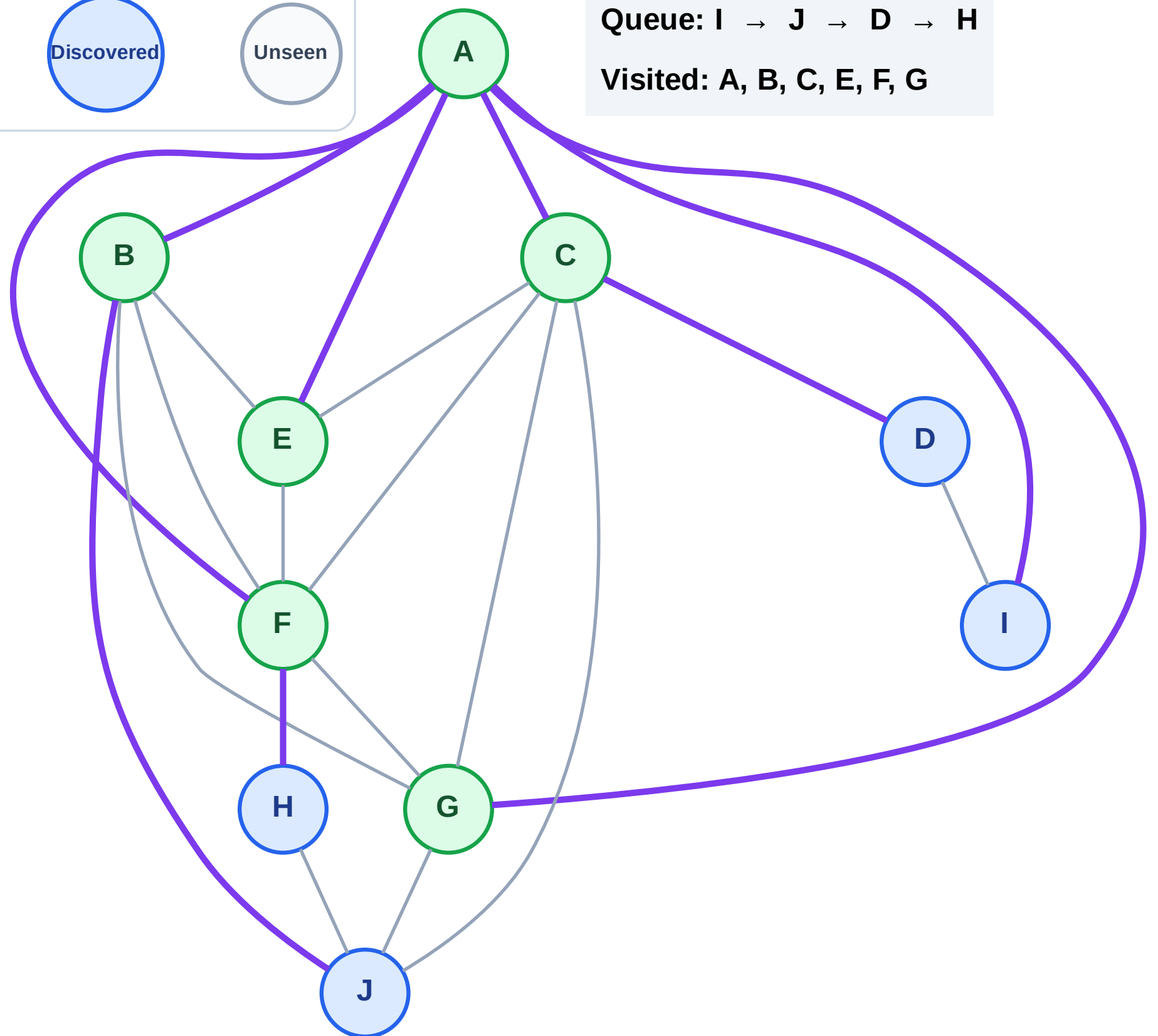
Step 13: No new neighbors from G

Legend



Queue: I → J → D → H

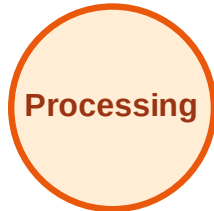
Visited: A, B, C, E, F, G



BFS Traversal

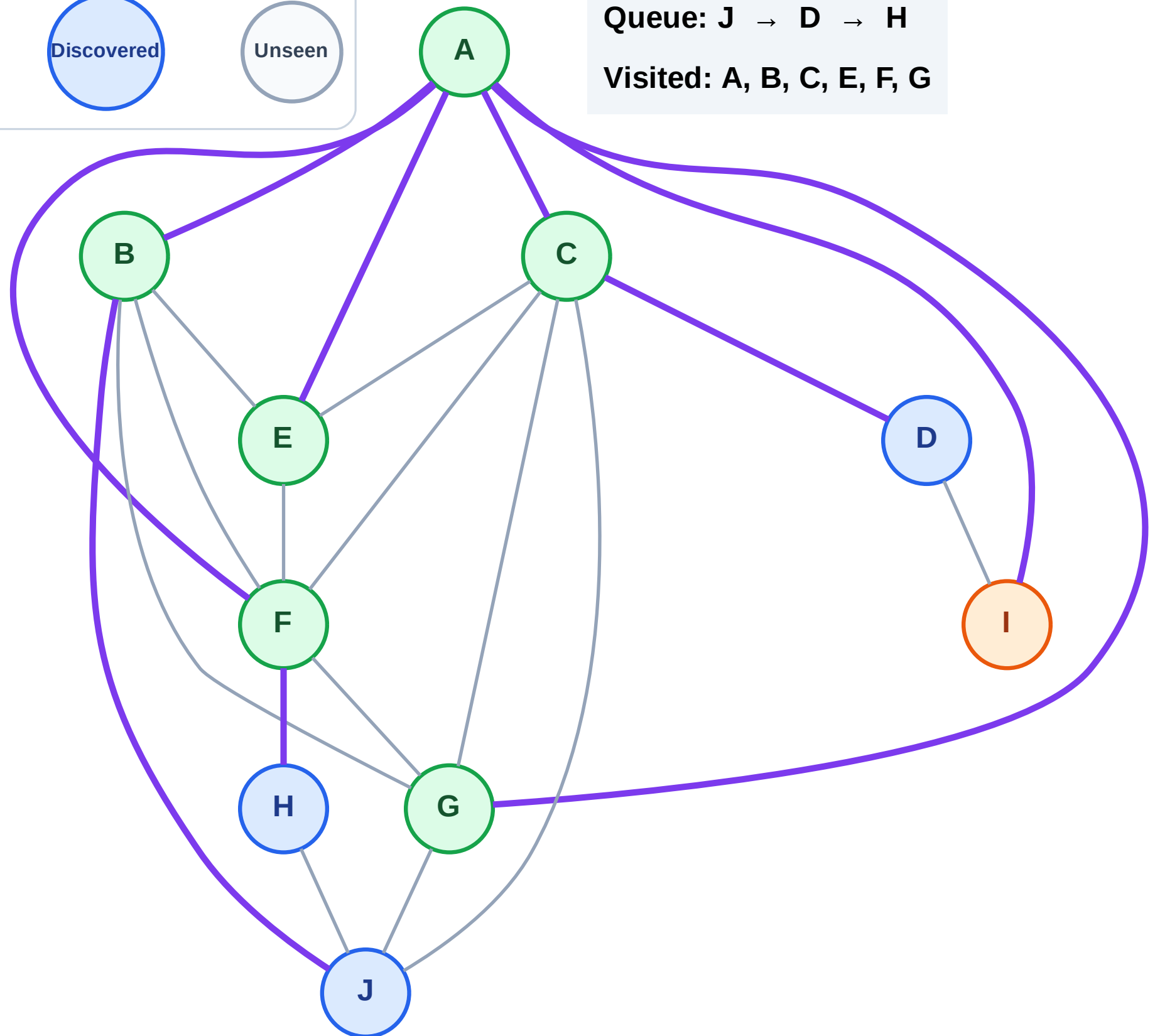
Step 14: Process node I

Legend



Queue: J → D → H

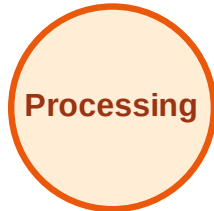
Visited: A, B, C, E, F, G



BFS Traversal

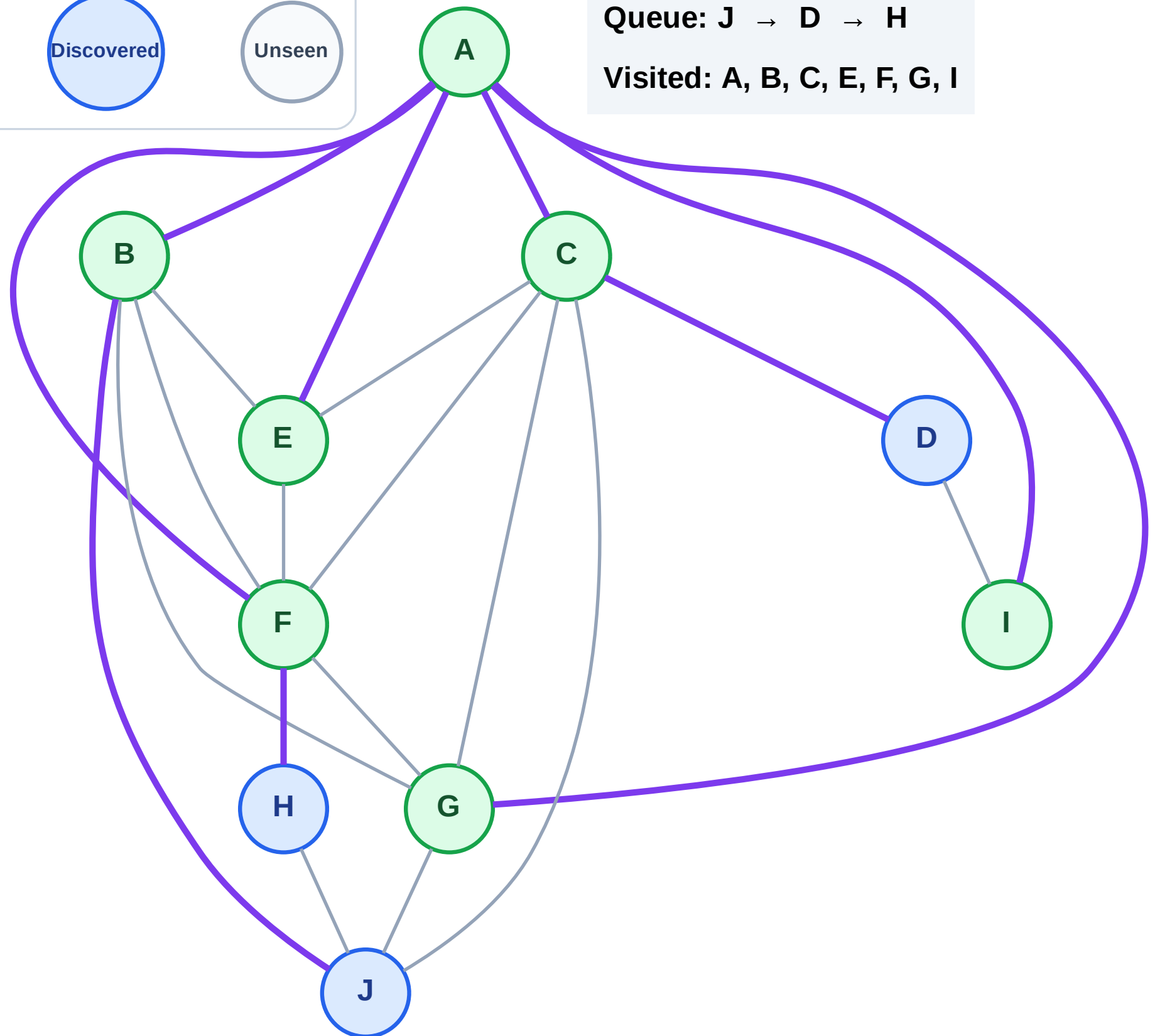
Step 15: No new neighbors from I

Legend



Queue: J → D → H

Visited: A, B, C, E, F, G, I



BFS Traversal

Step 16: Process node J

Legend

Complete

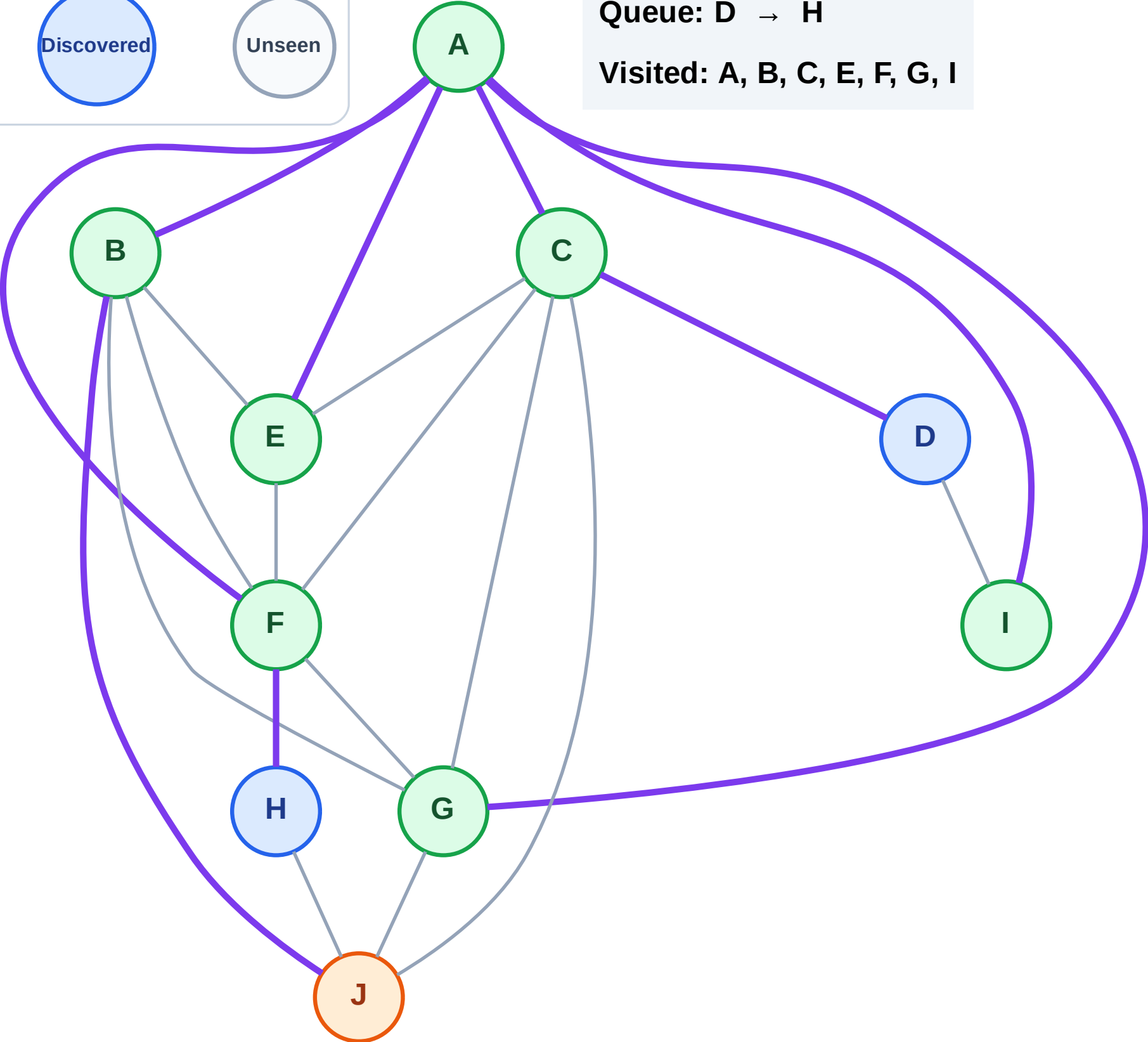
Processing

Discovered

Unseen

Queue: D → H

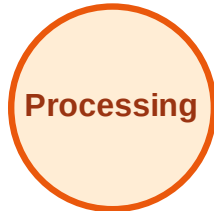
Visited: A, B, C, E, F, G, I



BFS Traversal

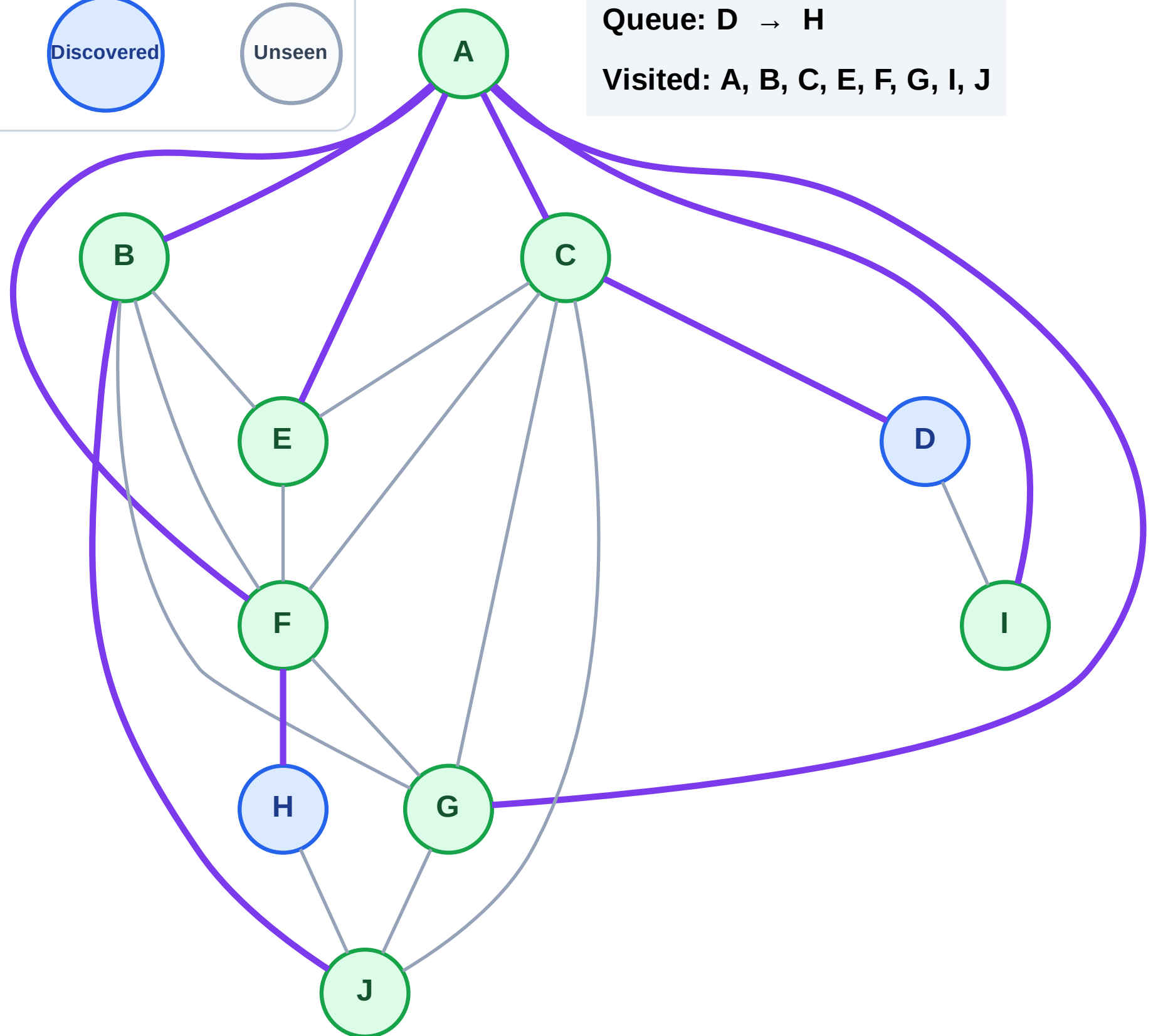
Step 17: No new neighbors from J

Legend



Queue: D → H

Visited: A, B, C, E, F, G, I, J



BFS Traversal

Step 18: Process node D

Legend



Complete



Processing



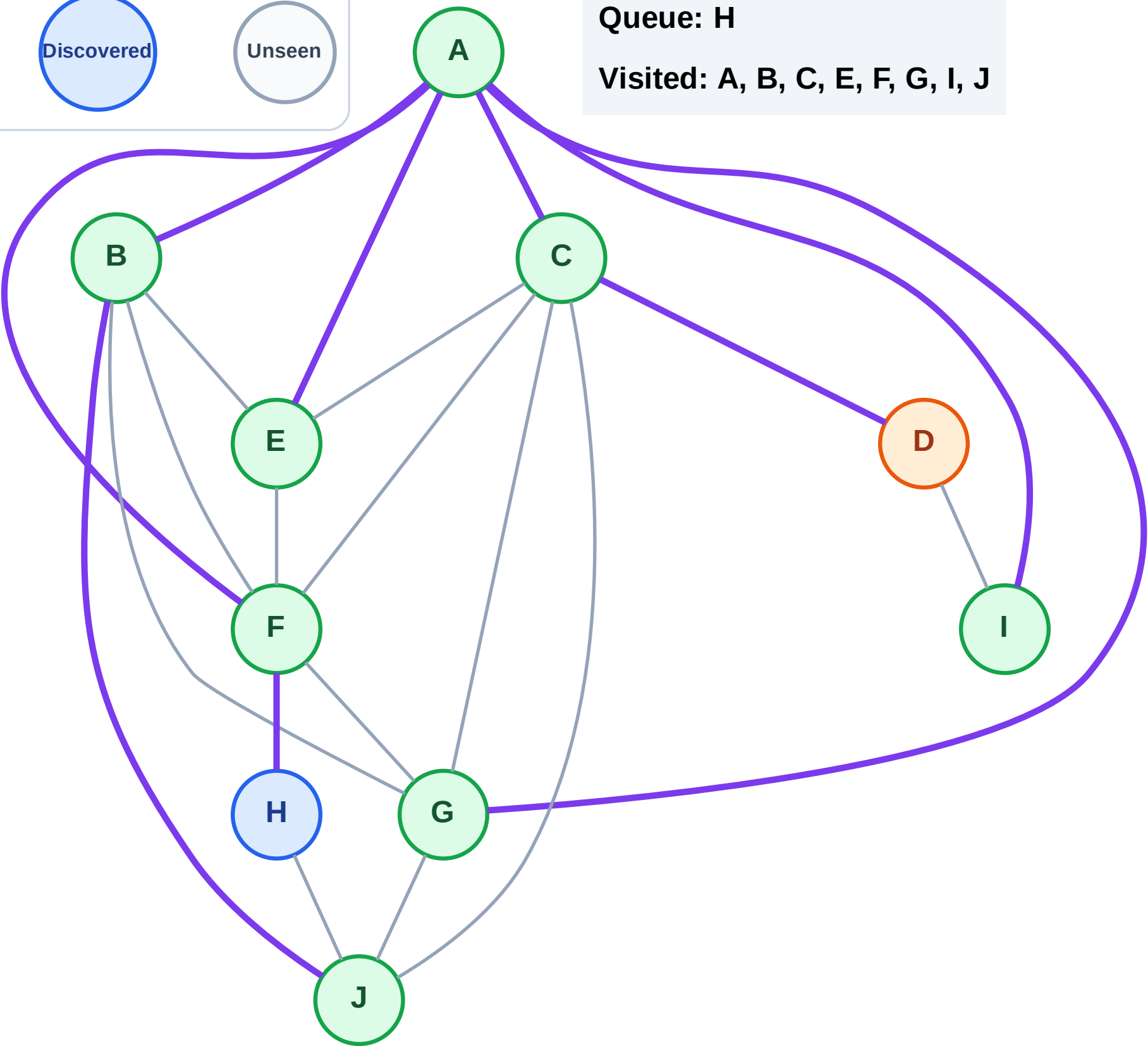
Discovered



Unseen

Queue: H

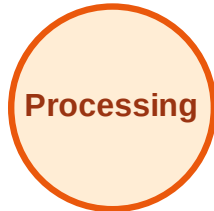
Visited: A, B, C, E, F, G, I, J



BFS Traversal

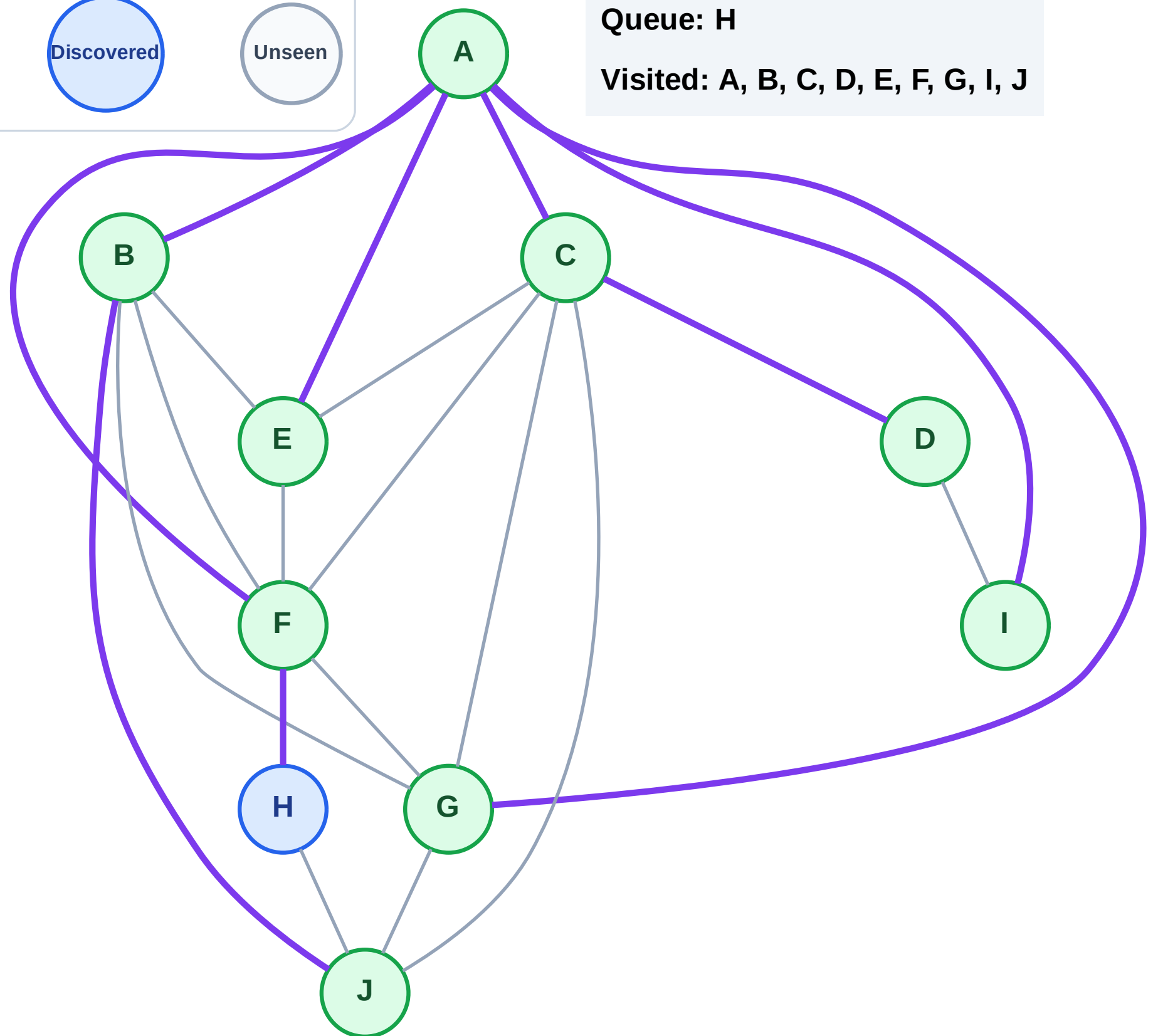
Step 19: No new neighbors from D

Legend



Queue: H

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

Step 20: Process node H

Legend

Complete

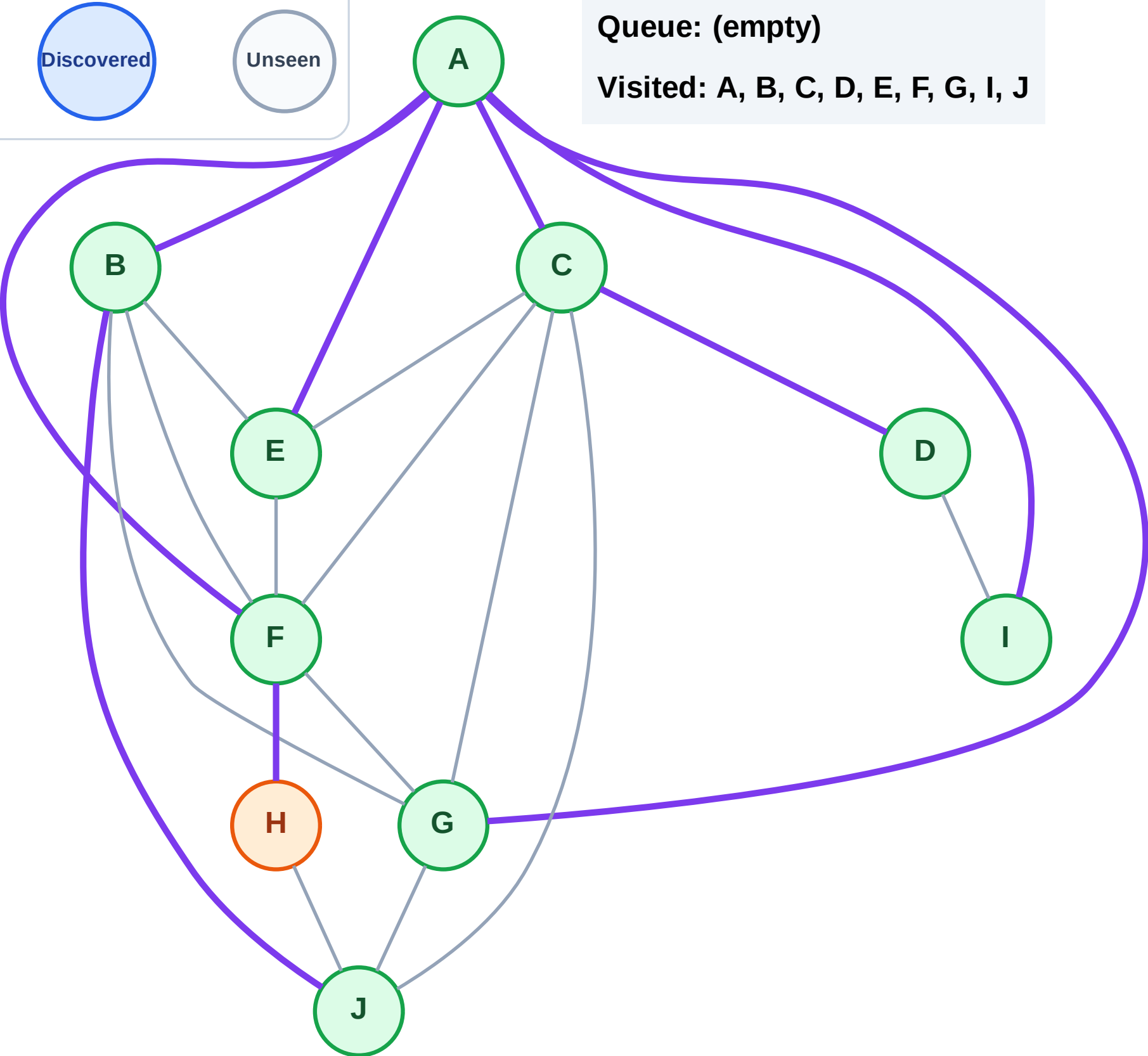
Processing

Discovered

Unseen

Queue: (empty)

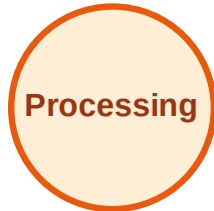
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

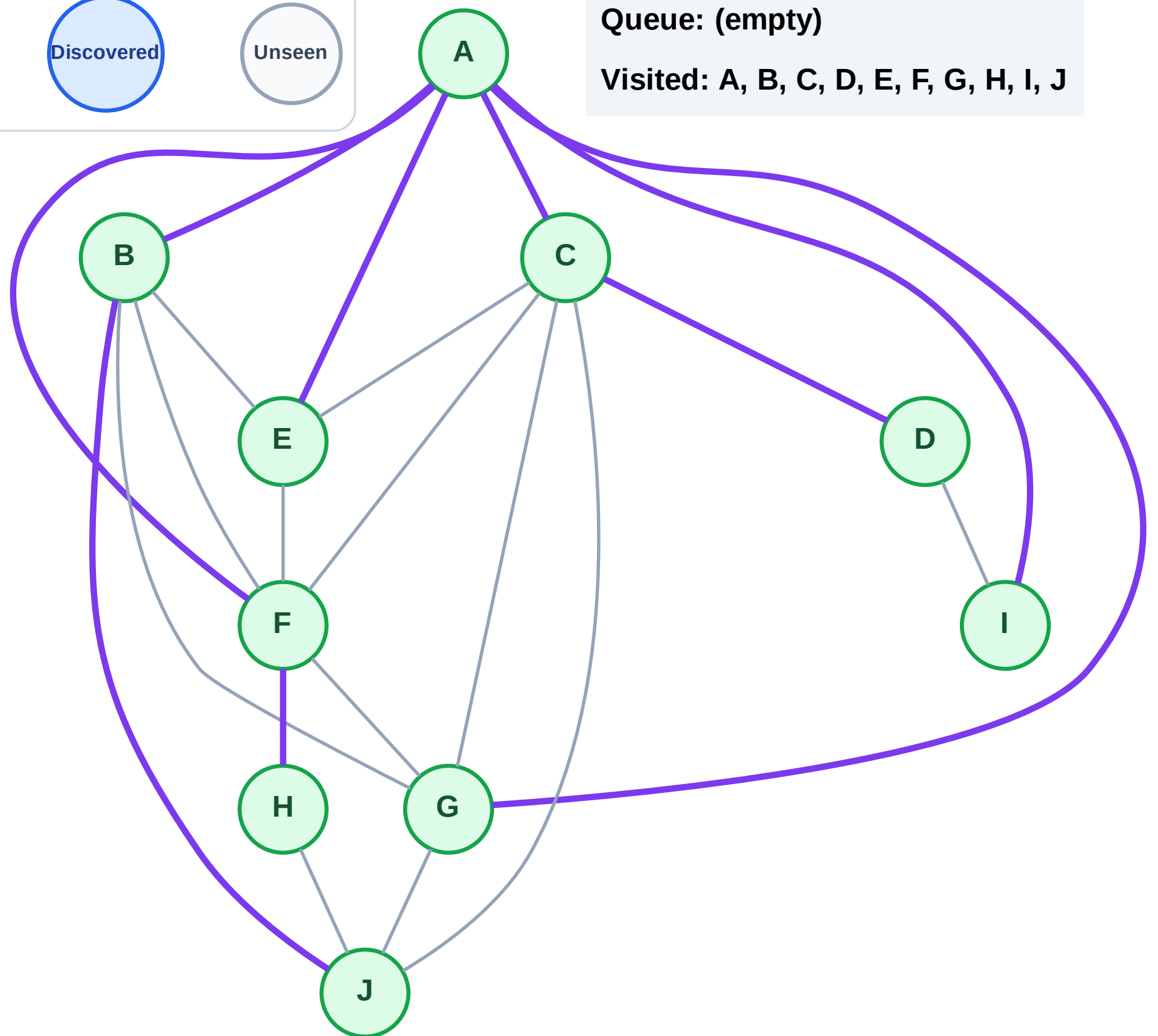
Step 21: No new neighbors from H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at A

Legend

Complete

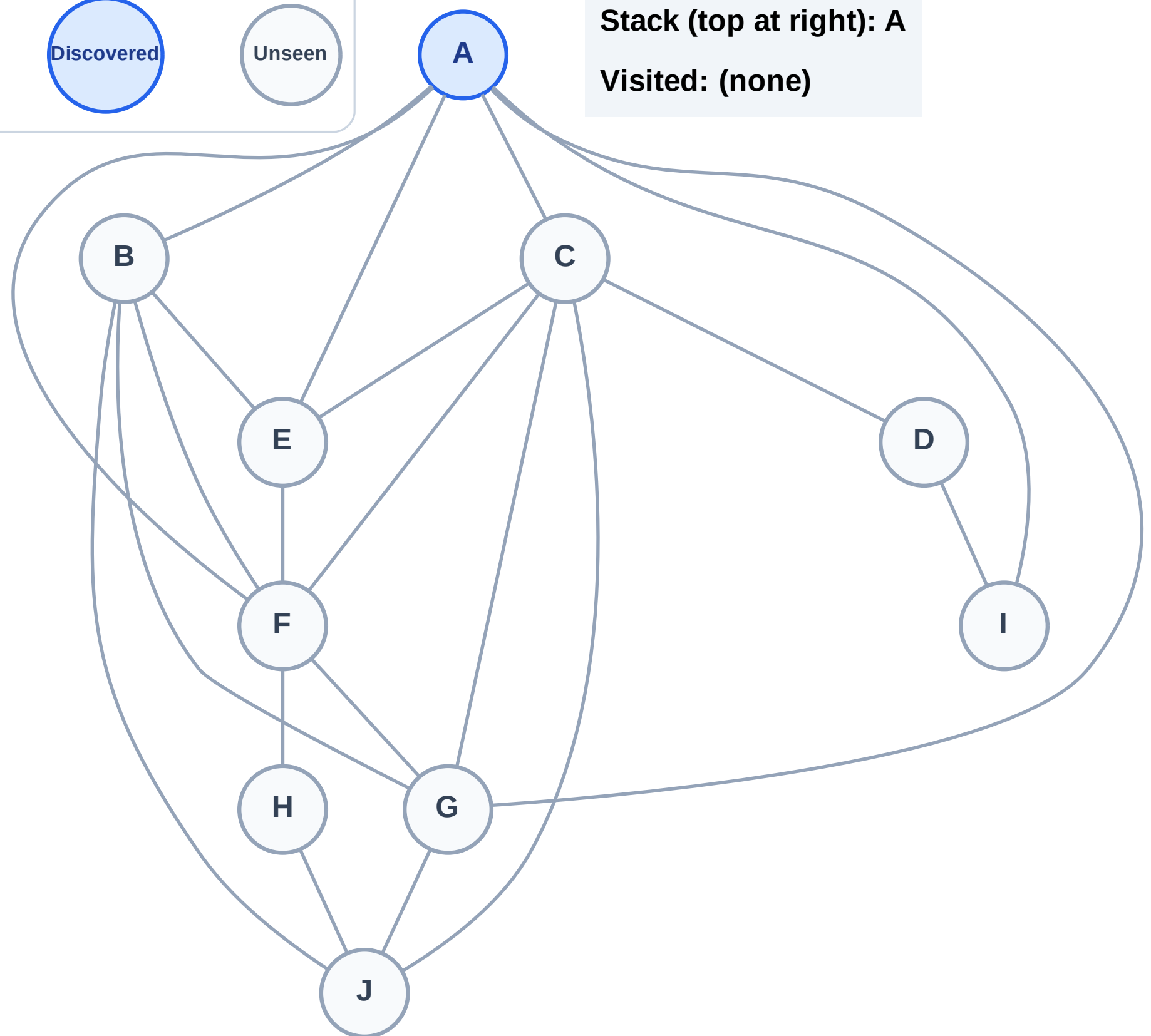
Processing

Discovered

Unseen

Stack (top at right): A

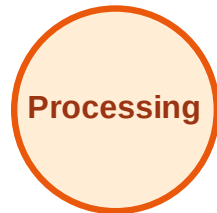
Visited: (none)



DFS Traversal

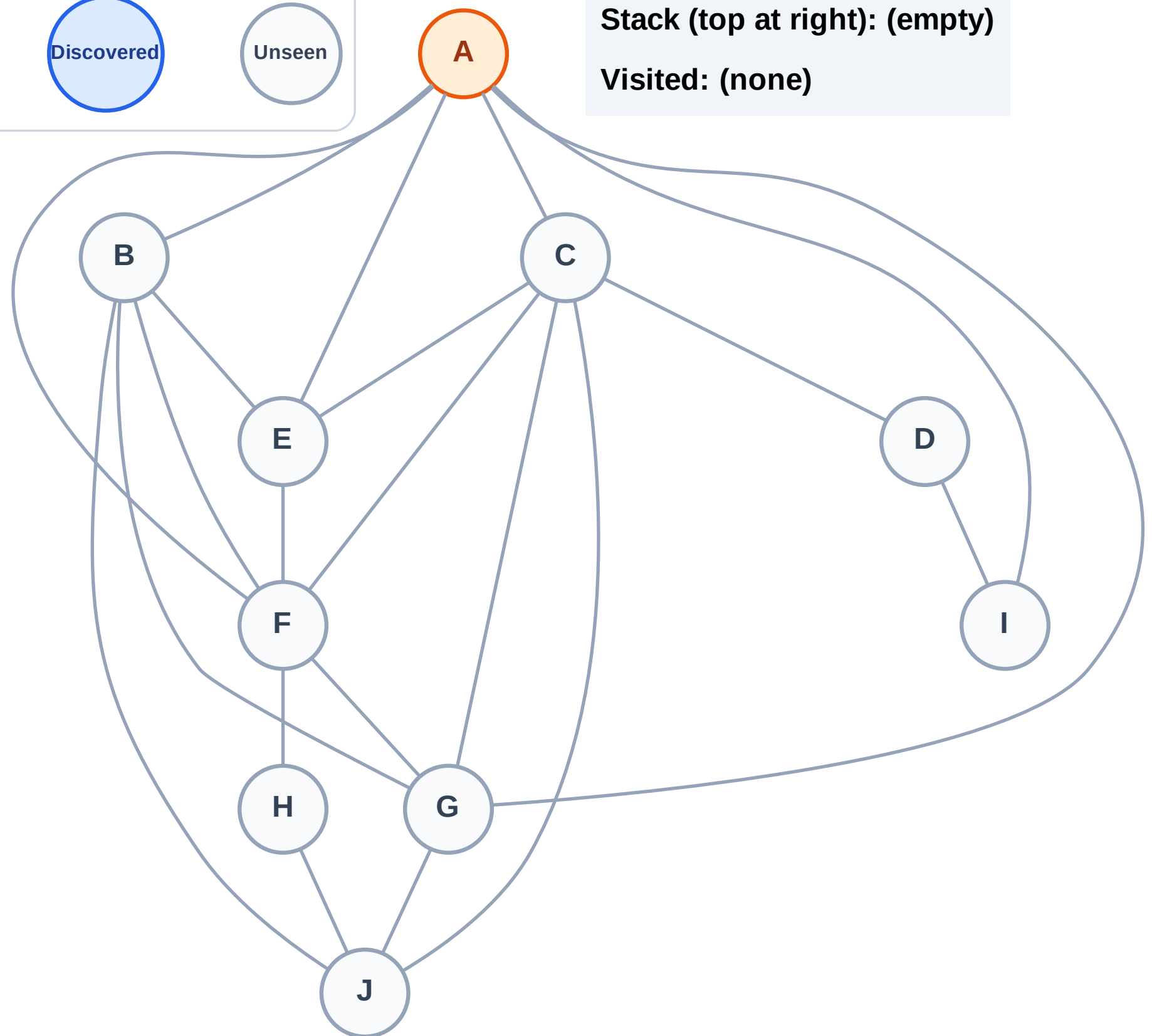
Step 2: Process node A

Legend



Stack (top at right): (empty)

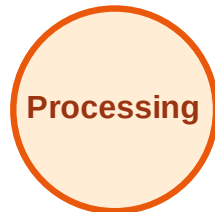
Visited: (none)



DFS Traversal

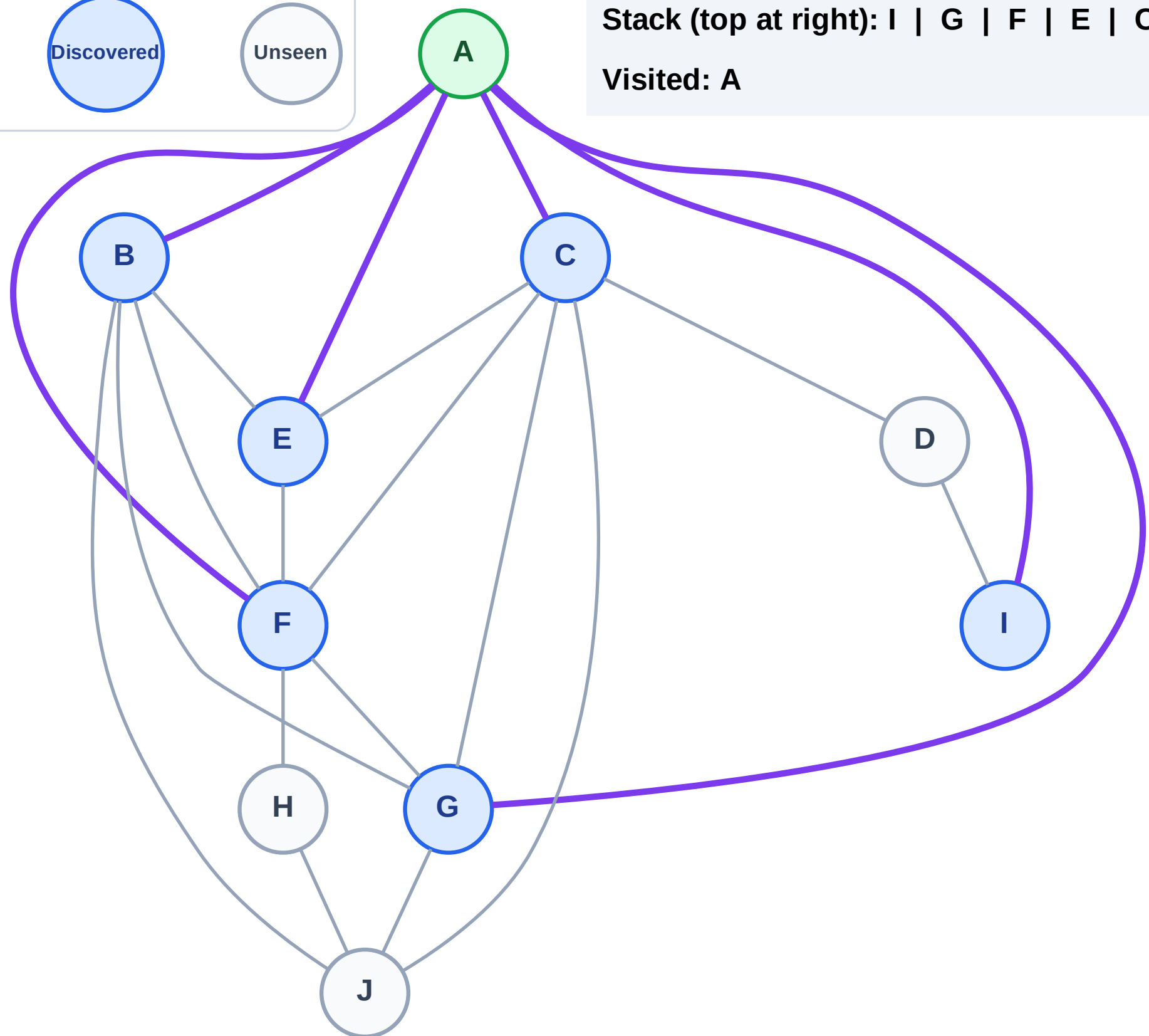
Step 3: Discover I, G, F, E, C, B from A

Legend



Stack (top at right): I | G | F | E | C | B

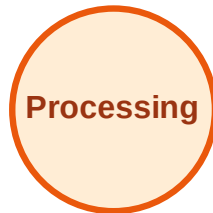
Visited: A



DFS Traversal

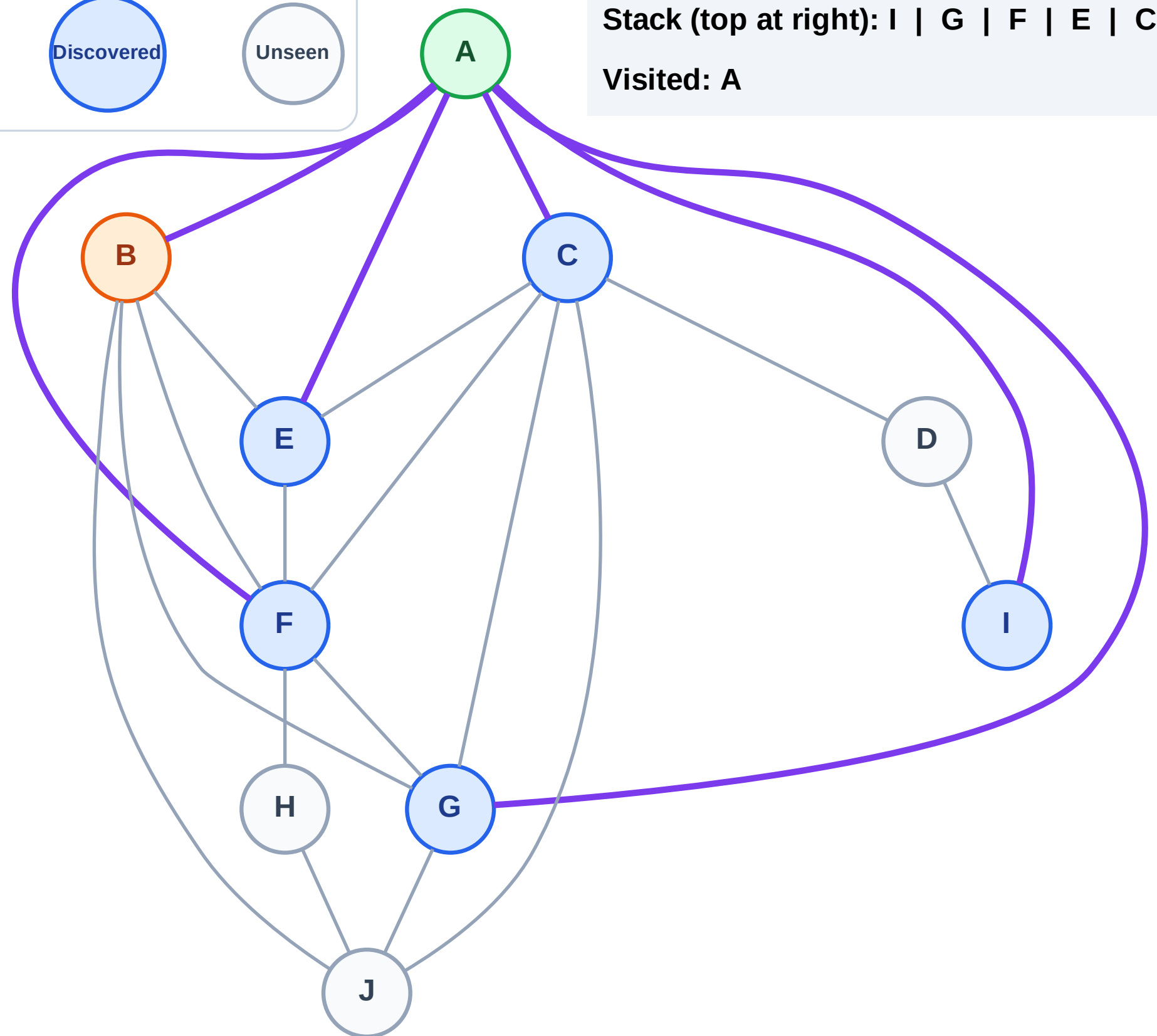
Step 4: Process node B

Legend



Stack (top at right): I | G | F | E | C

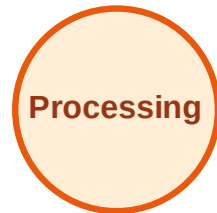
Visited: A



DFS Traversal

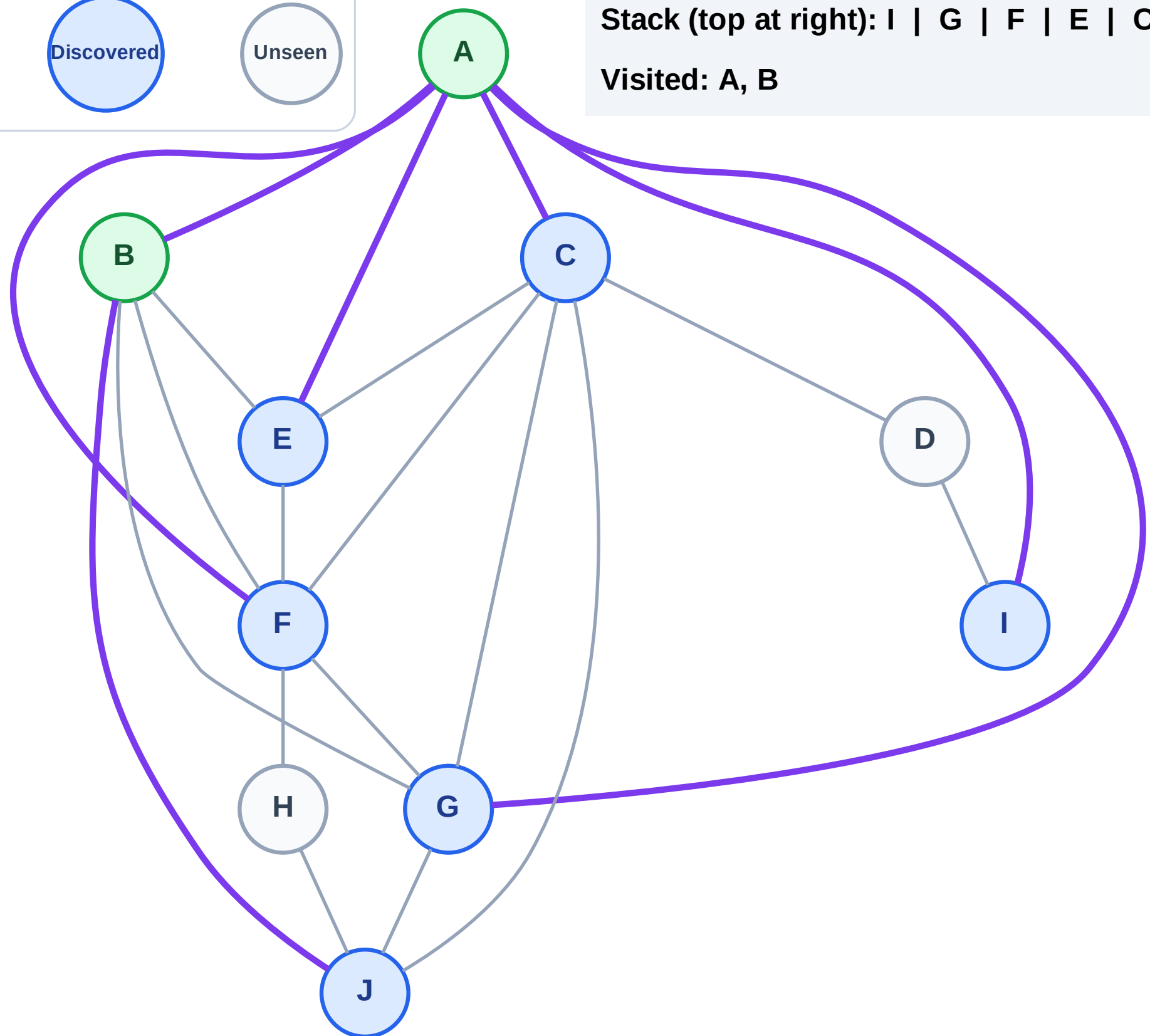
Step 5: Discover J from B

Legend



Stack (top at right): I | G | F | E | C | J

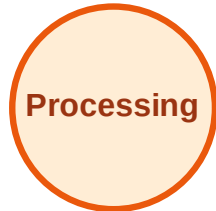
Visited: A, B



DFS Traversal

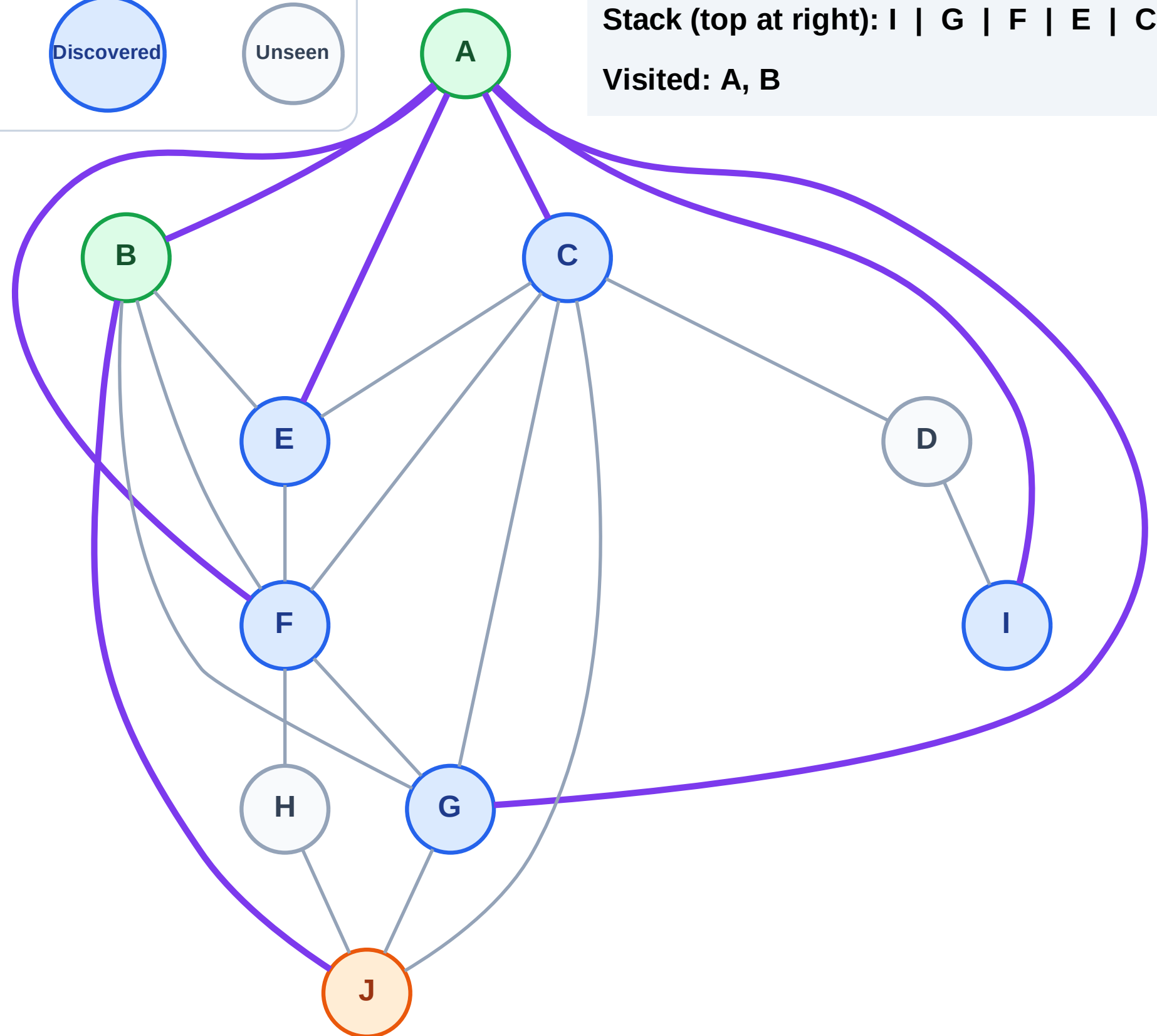
Step 6: Process node J

Legend



Stack (top at right): I | G | F | E | C

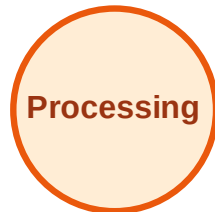
Visited: A, B



DFS Traversal

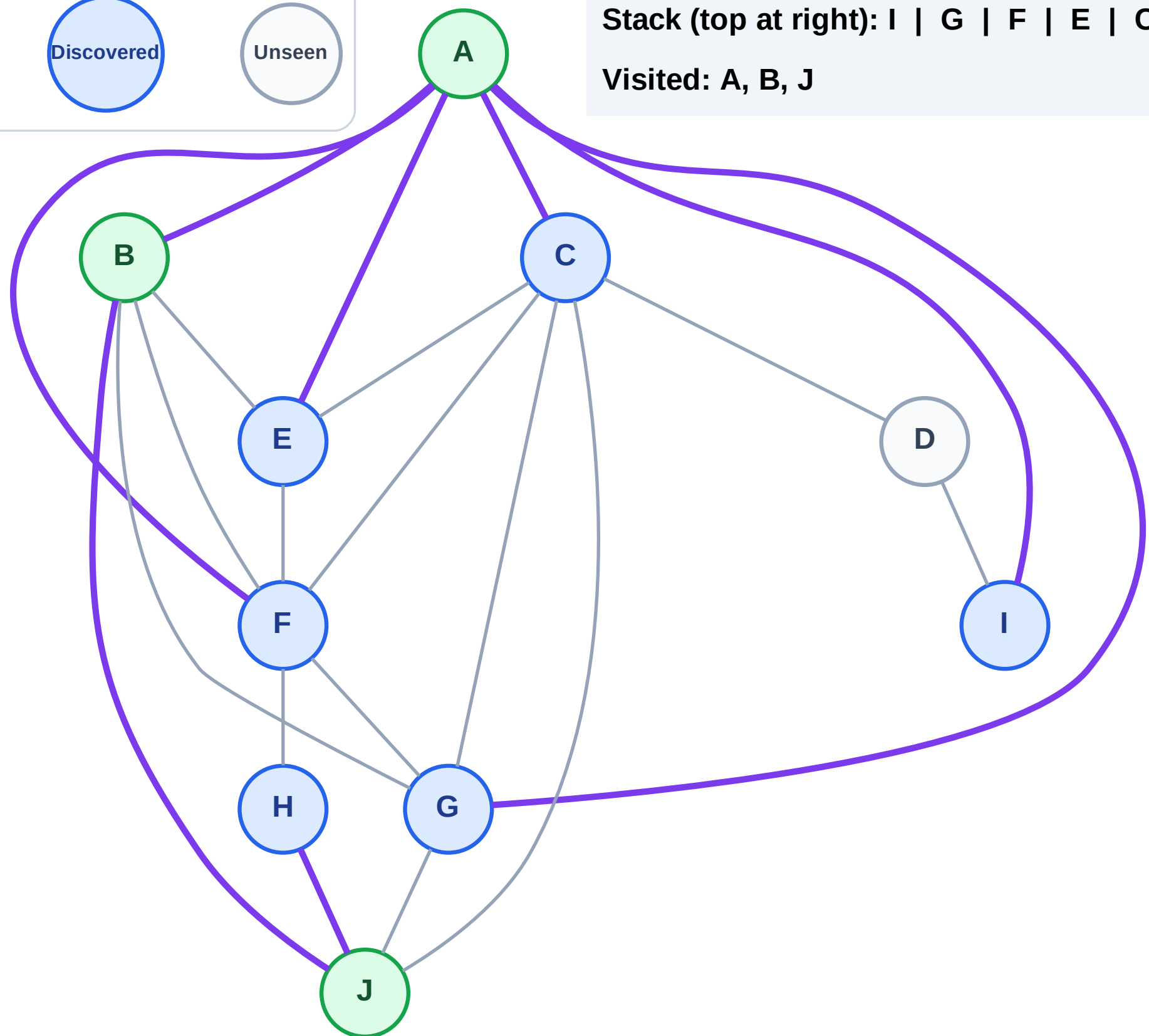
Step 7: Discover H from J

Legend



Stack (top at right): I | G | F | E | C | H

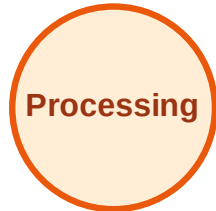
Visited: A, B, J



DFS Traversal

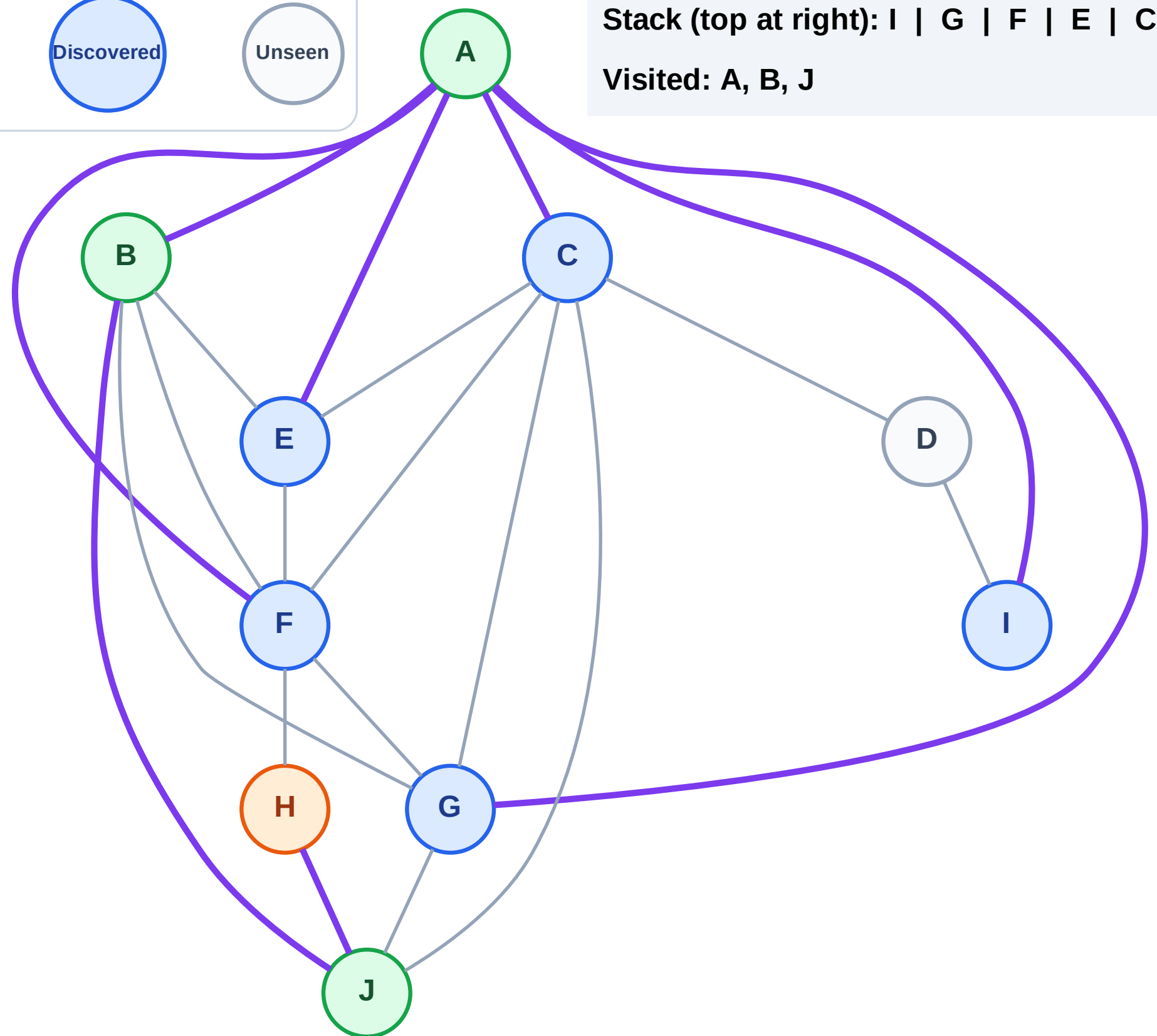
Step 8: Process node H

Legend



Stack (top at right): I | G | F | E | C

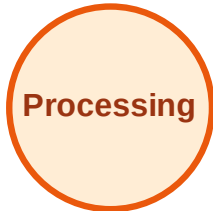
Visited: A, B, J



DFS Traversal

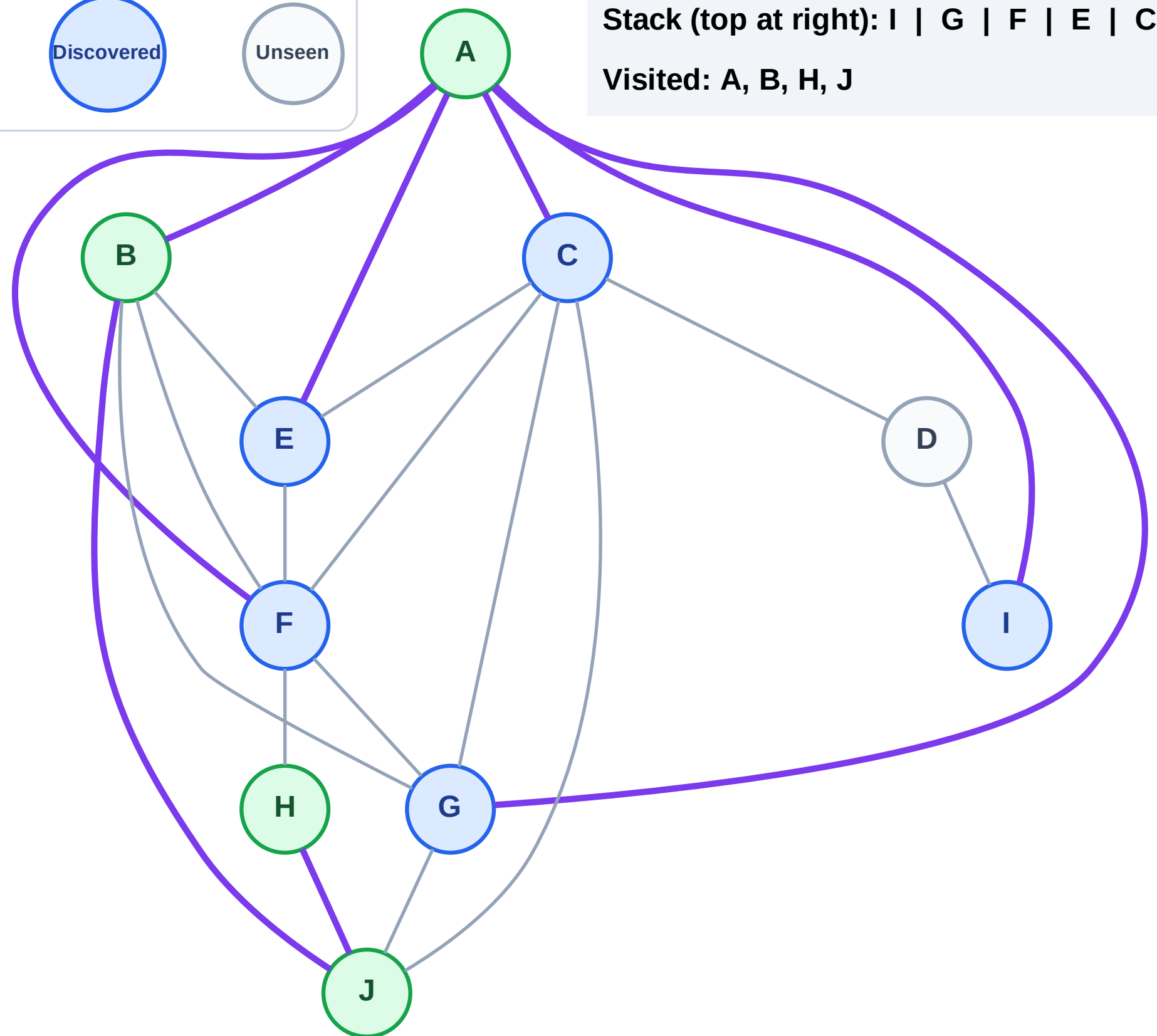
Step 9: No new neighbors from H

Legend



Stack (top at right): I | G | F | E | C

Visited: A, B, H, J



DFS Traversal

Step 10: Process node C

Legend

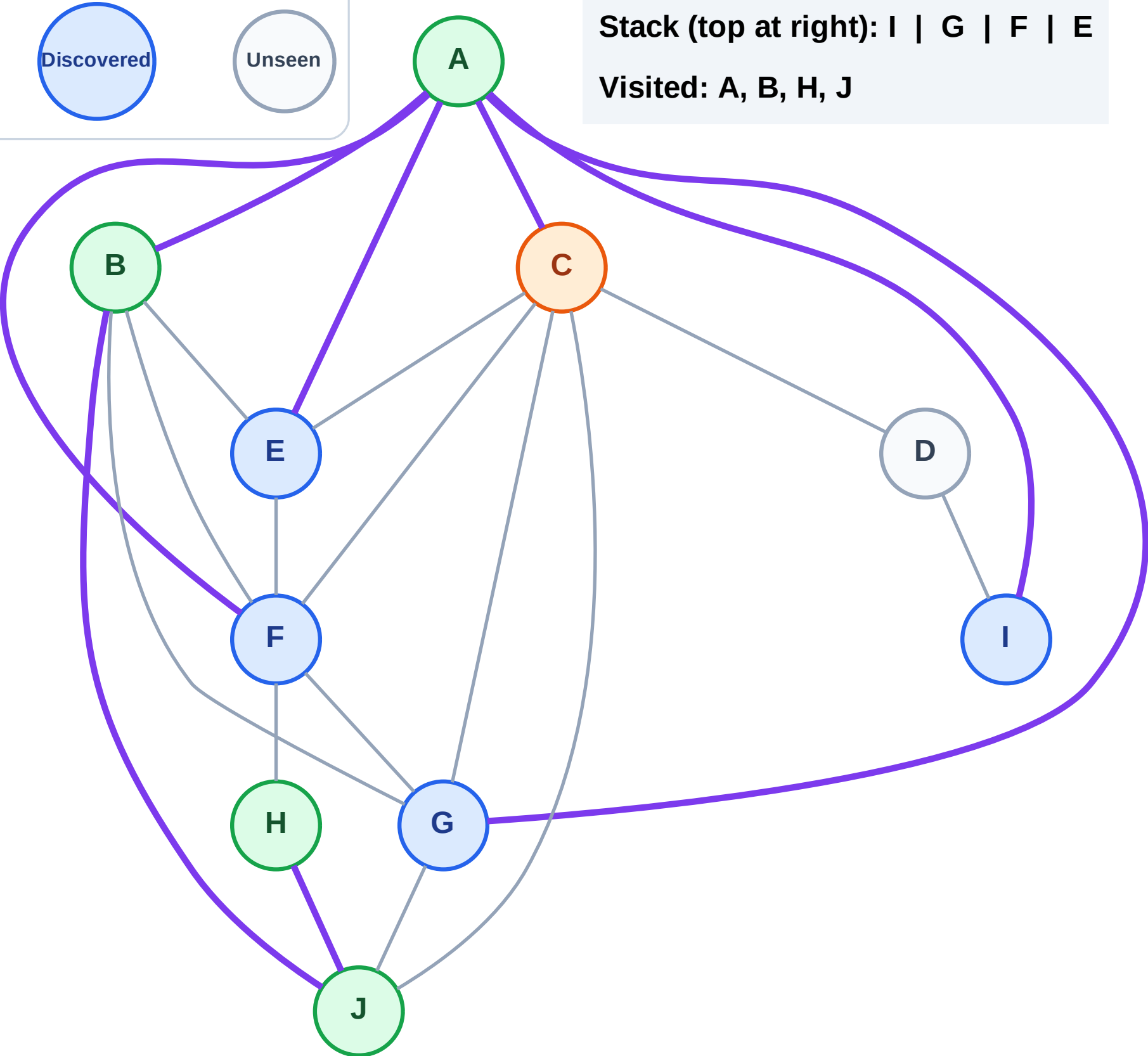
Complete

Processing

Discovered

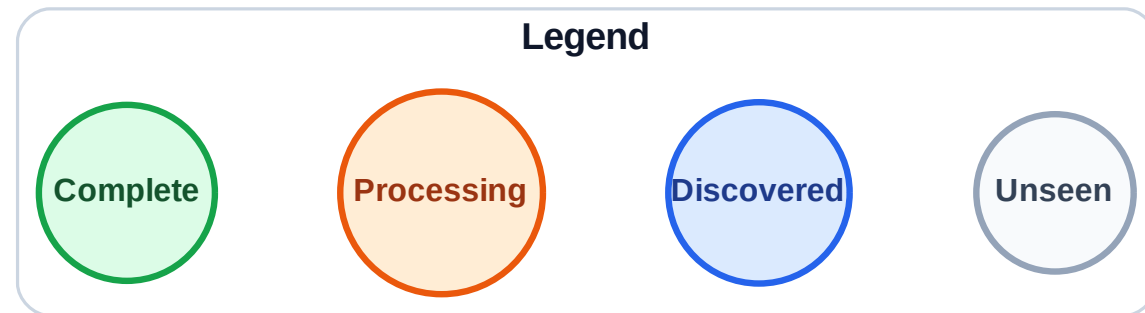
Unseen

Stack (top at right): I | G | F | E
Visited: A, B, H, J



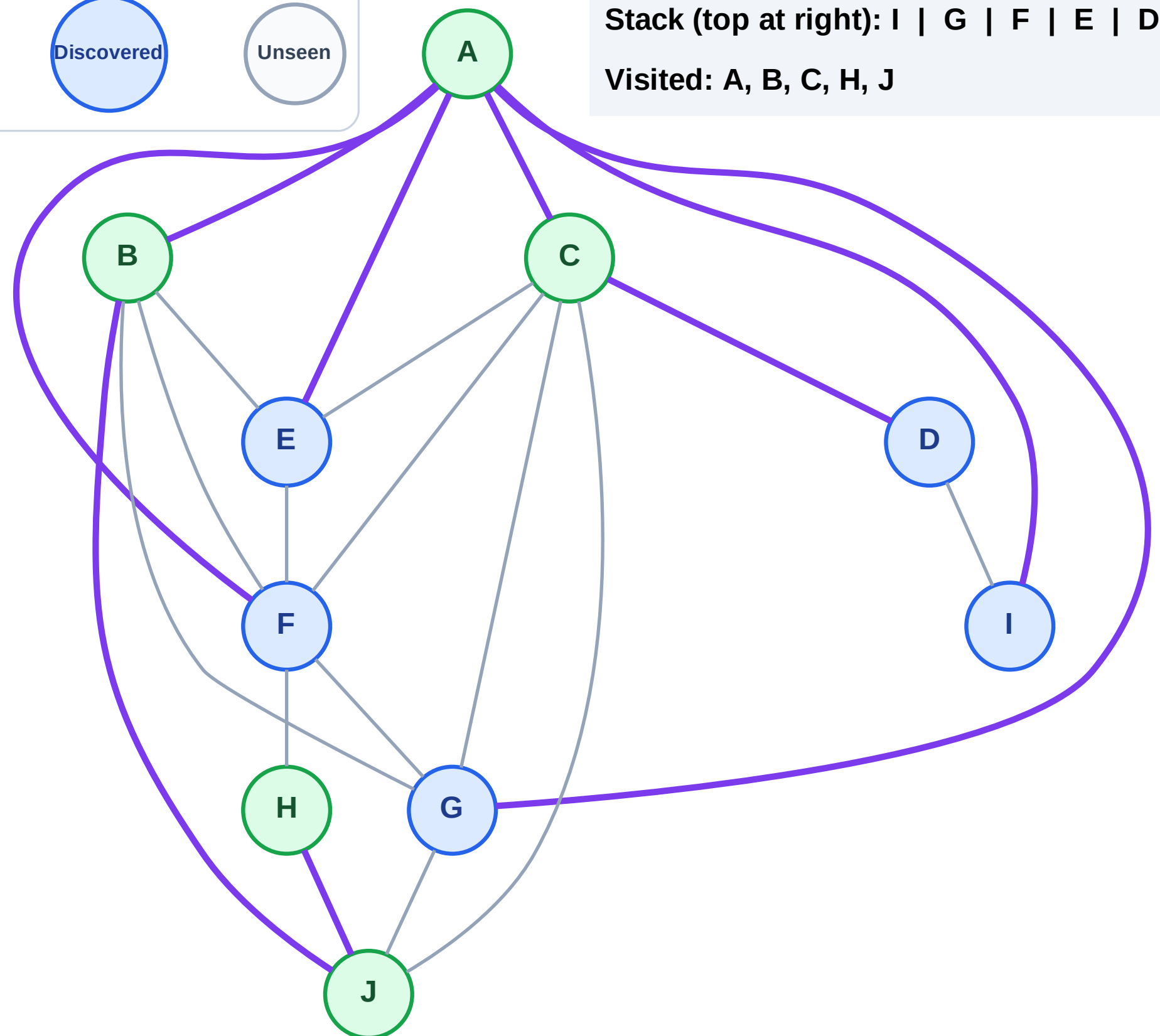
DFS Traversal

Step 11: Discover D from C



Stack (top at right): I | G | F | E | D

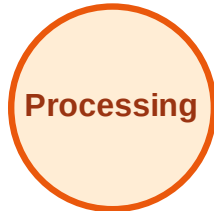
Visited: A, B, C, H, J



DFS Traversal

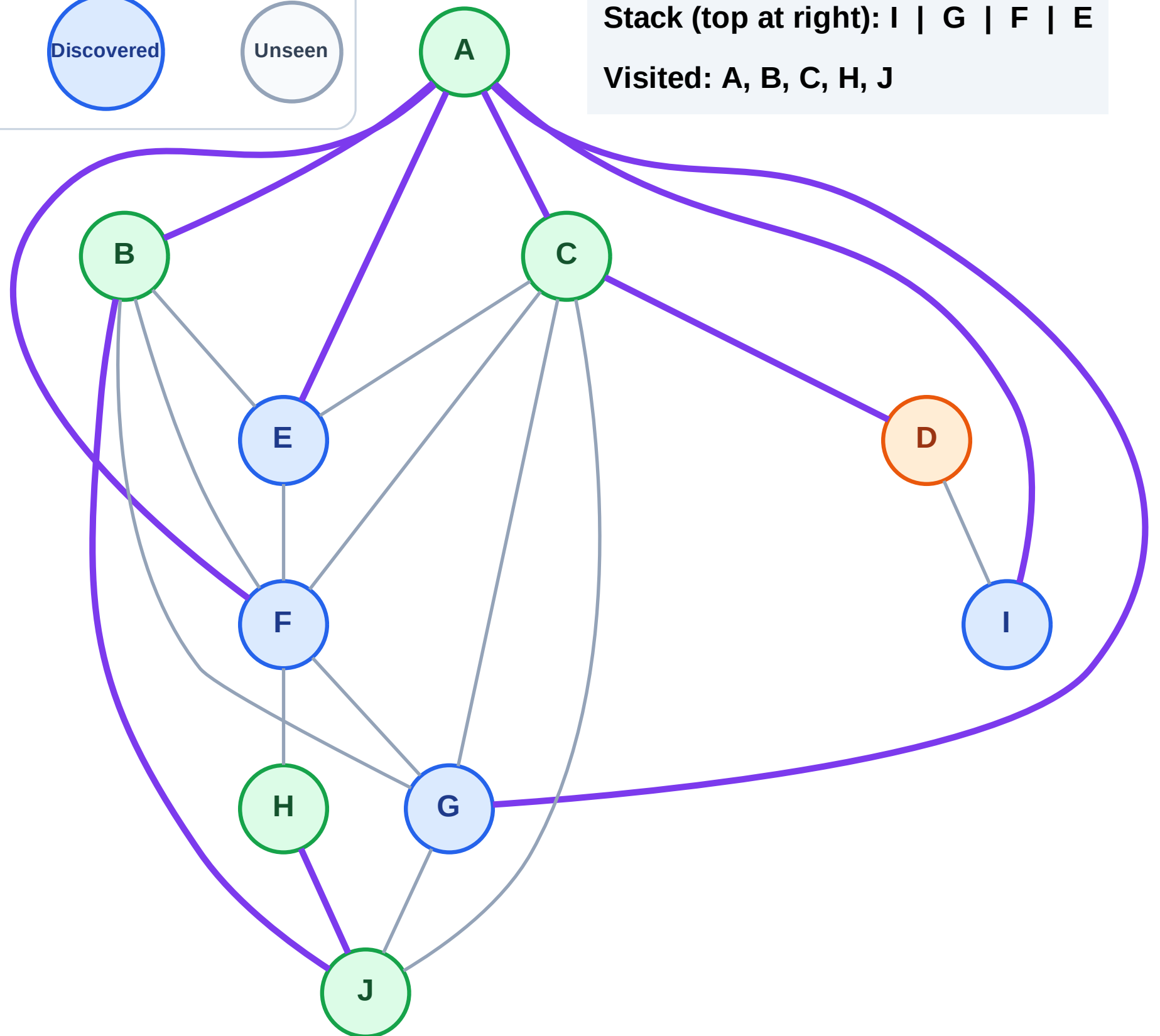
Step 12: Process node D

Legend



Stack (top at right): I | G | F | E

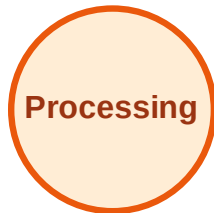
Visited: A, B, C, H, J



DFS Traversal

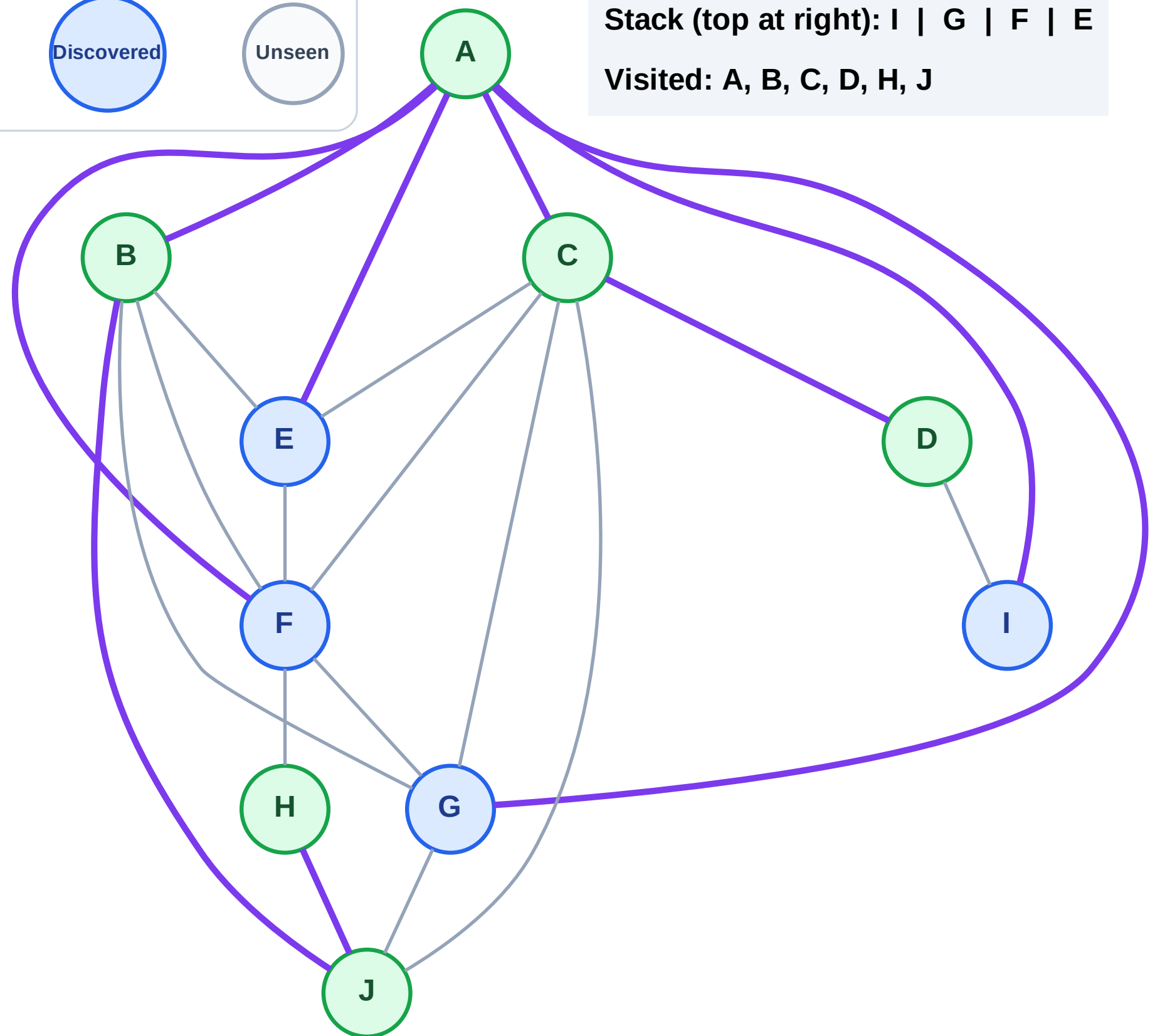
Step 13: No new neighbors from D

Legend



Stack (top at right): I | G | F | E

Visited: A, B, C, D, H, J



DFS Traversal

Step 14: Process node E

Legend

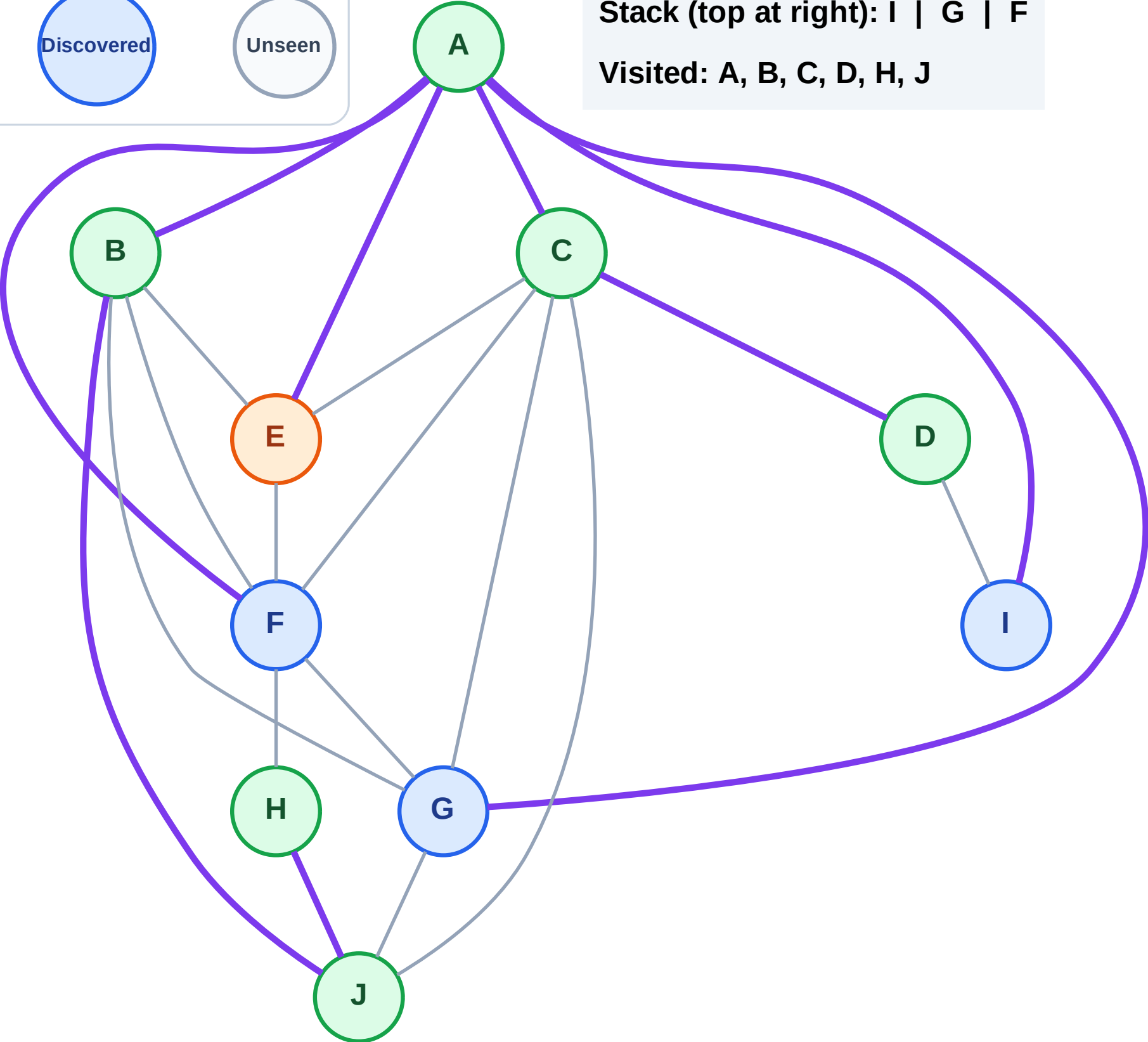
Complete

Processing

Discovered

Unseen

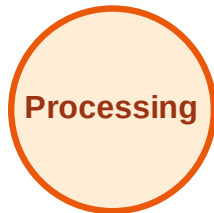
Stack (top at right): I | G | F
Visited: A, B, C, D, H, J



DFS Traversal

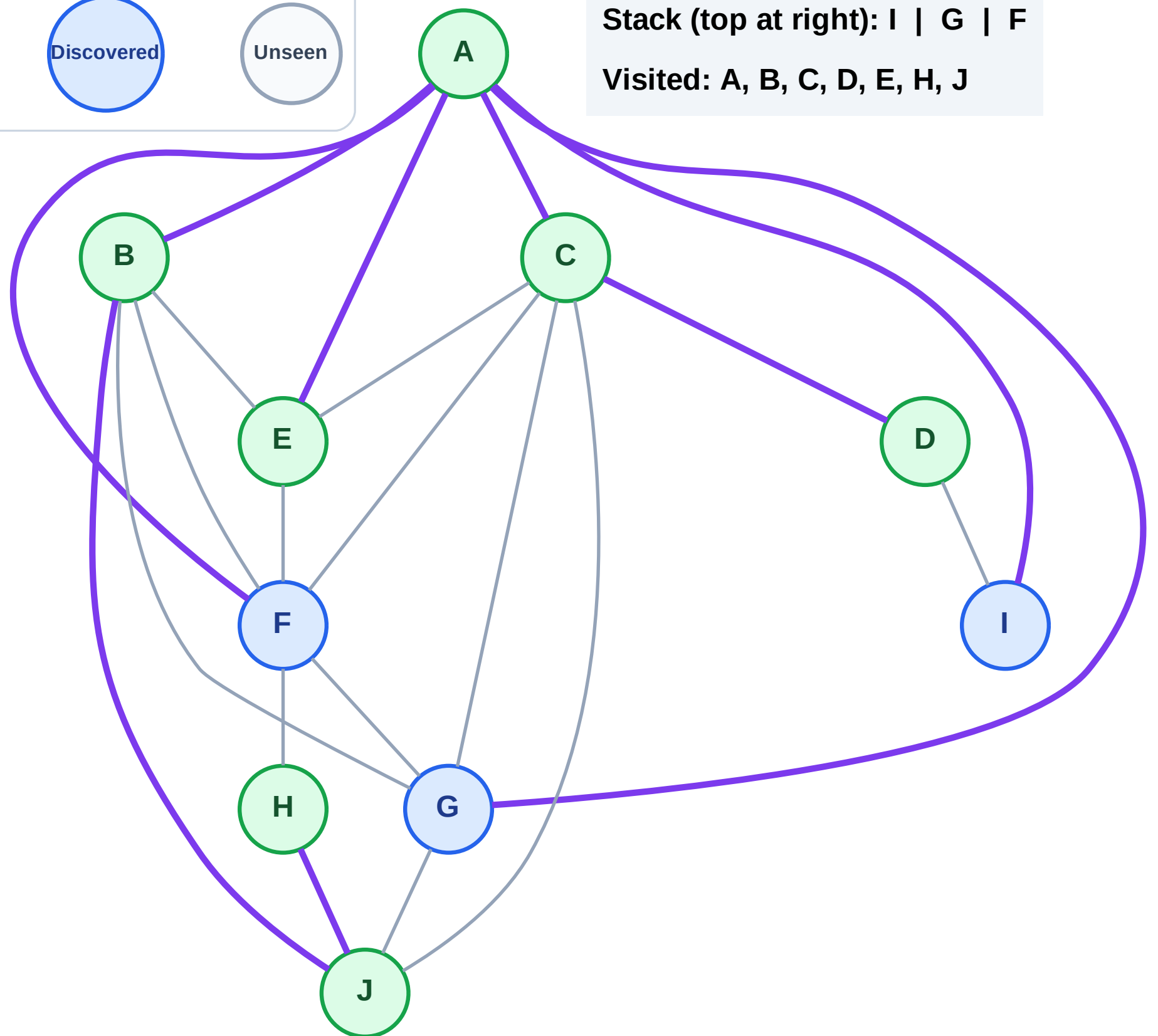
Step 15: No new neighbors from E

Legend



Stack (top at right): I | G | F

Visited: A, B, C, D, E, H, J



DFS Traversal

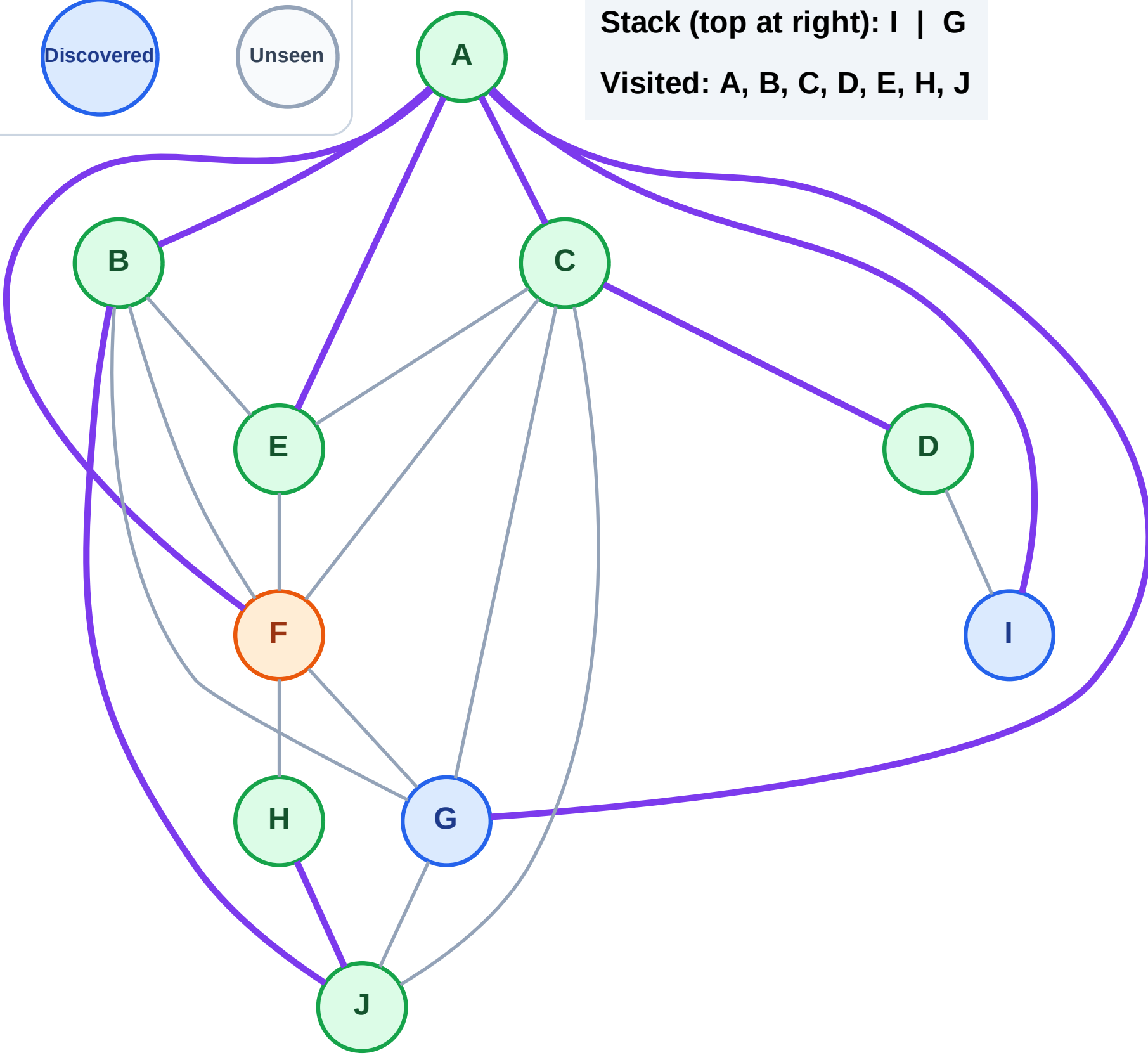
Step 16: Process node F

Legend



Stack (top at right): I | G

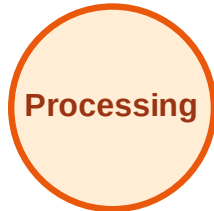
Visited: A, B, C, D, E, H, J



DFS Traversal

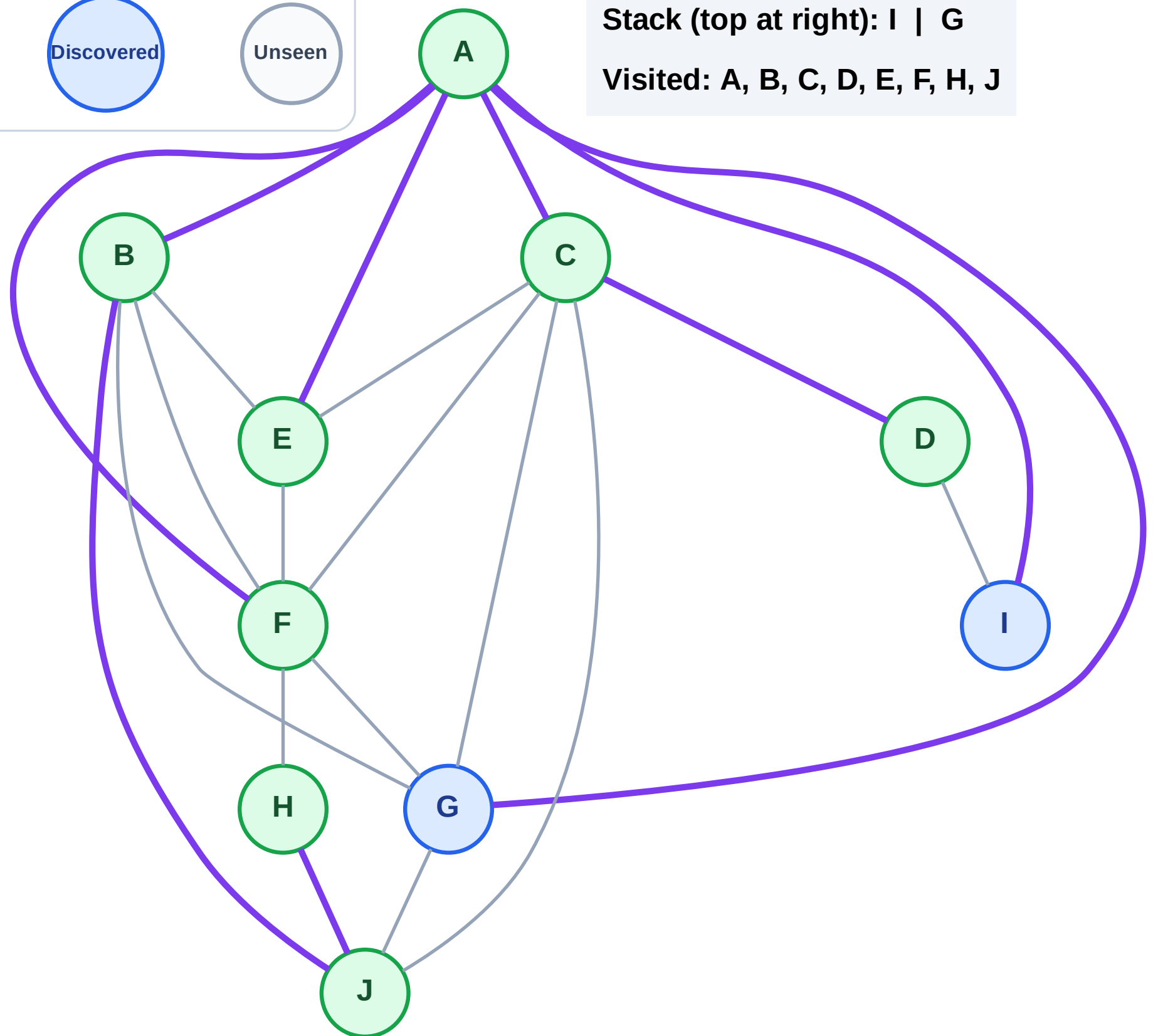
Step 17: No new neighbors from F

Legend



Stack (top at right): I | G

Visited: A, B, C, D, E, F, H, J



DFS Traversal

Step 18: Process node G

Legend

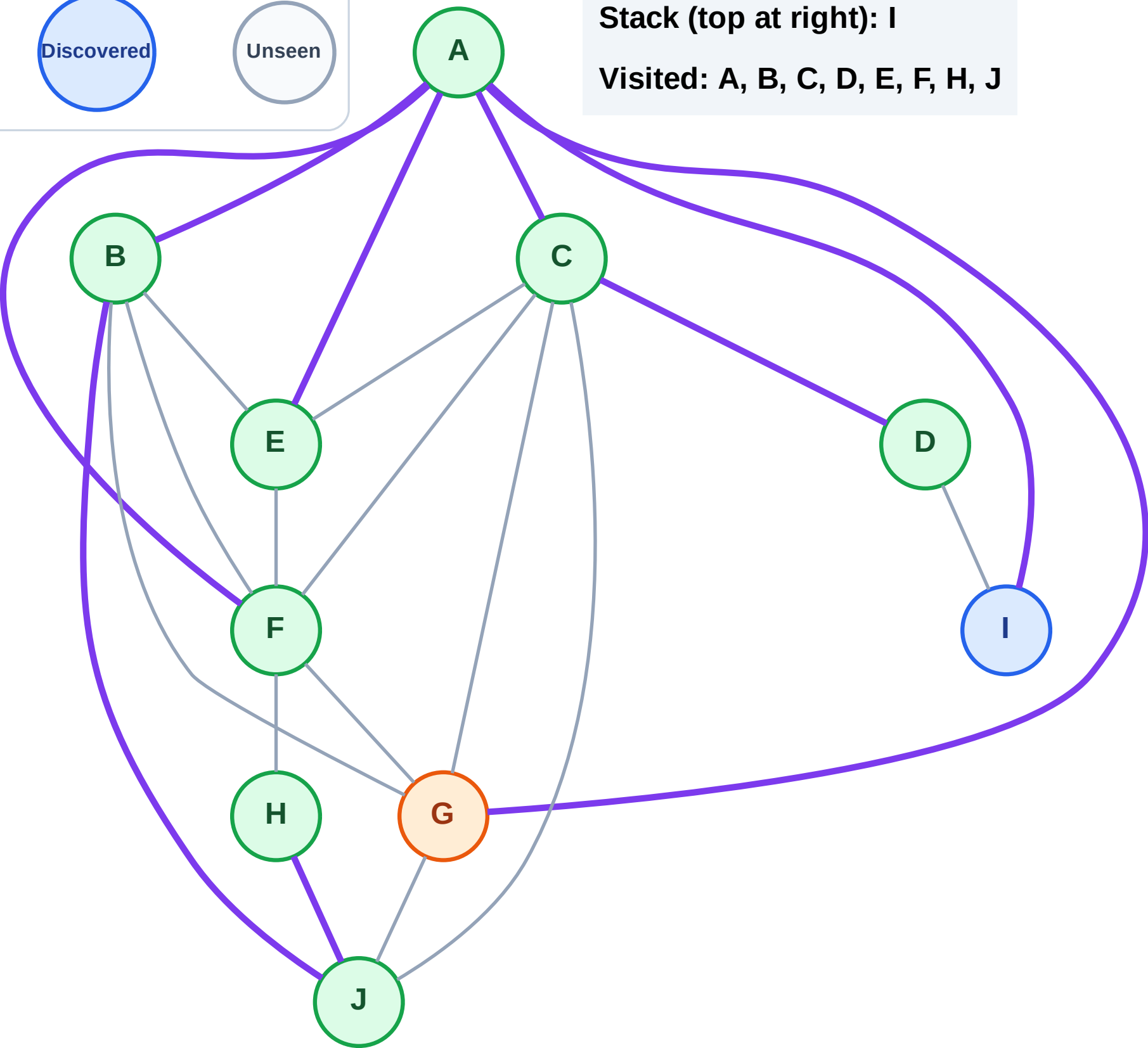
Complete

Processing

Discovered

Unseen

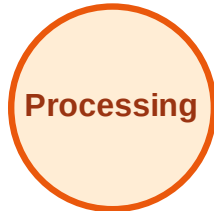
Stack (top at right): I
Visited: A, B, C, D, E, F, H, J



DFS Traversal

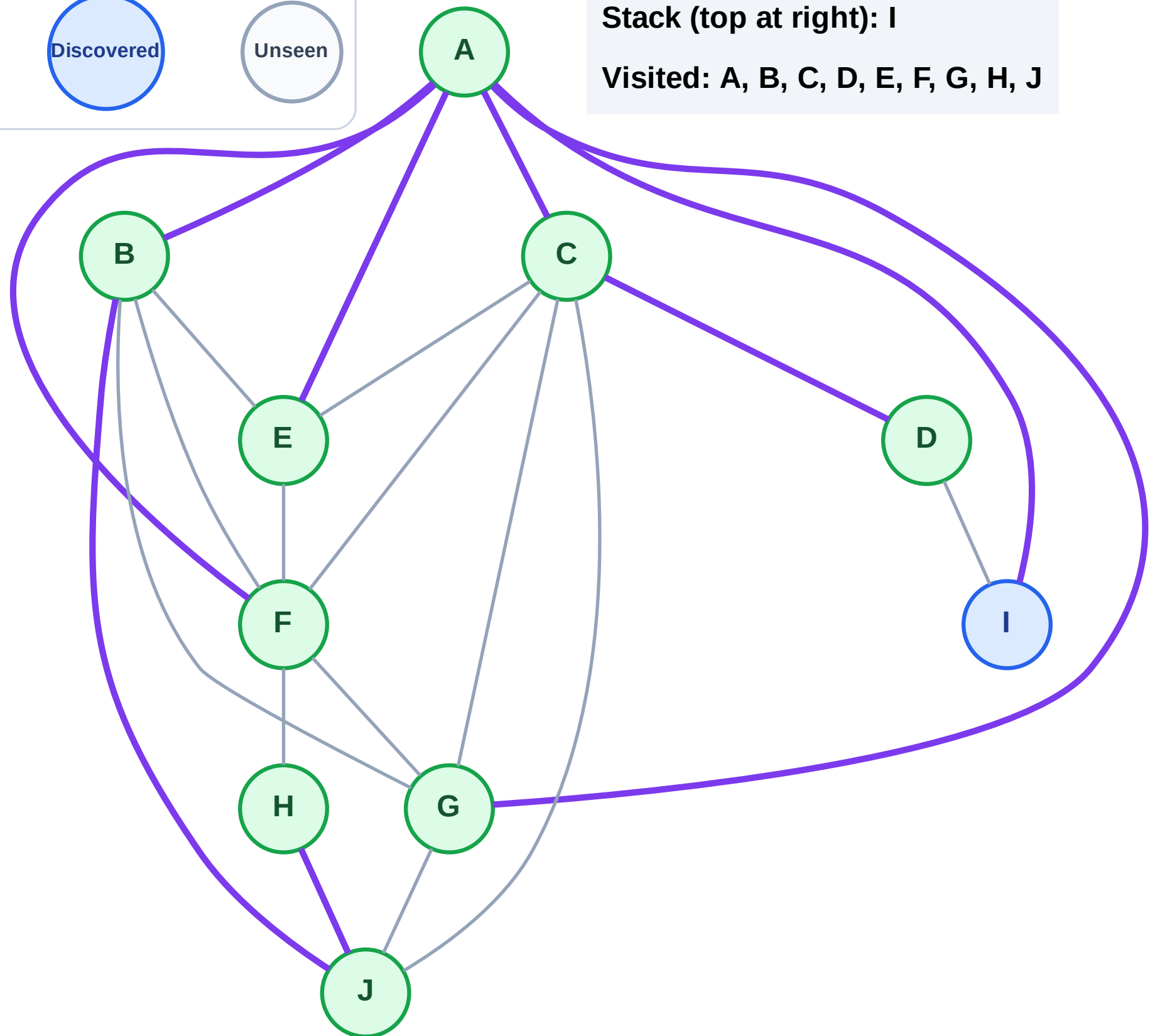
Step 19: No new neighbors from G

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 20: Process node I

Legend

Complete

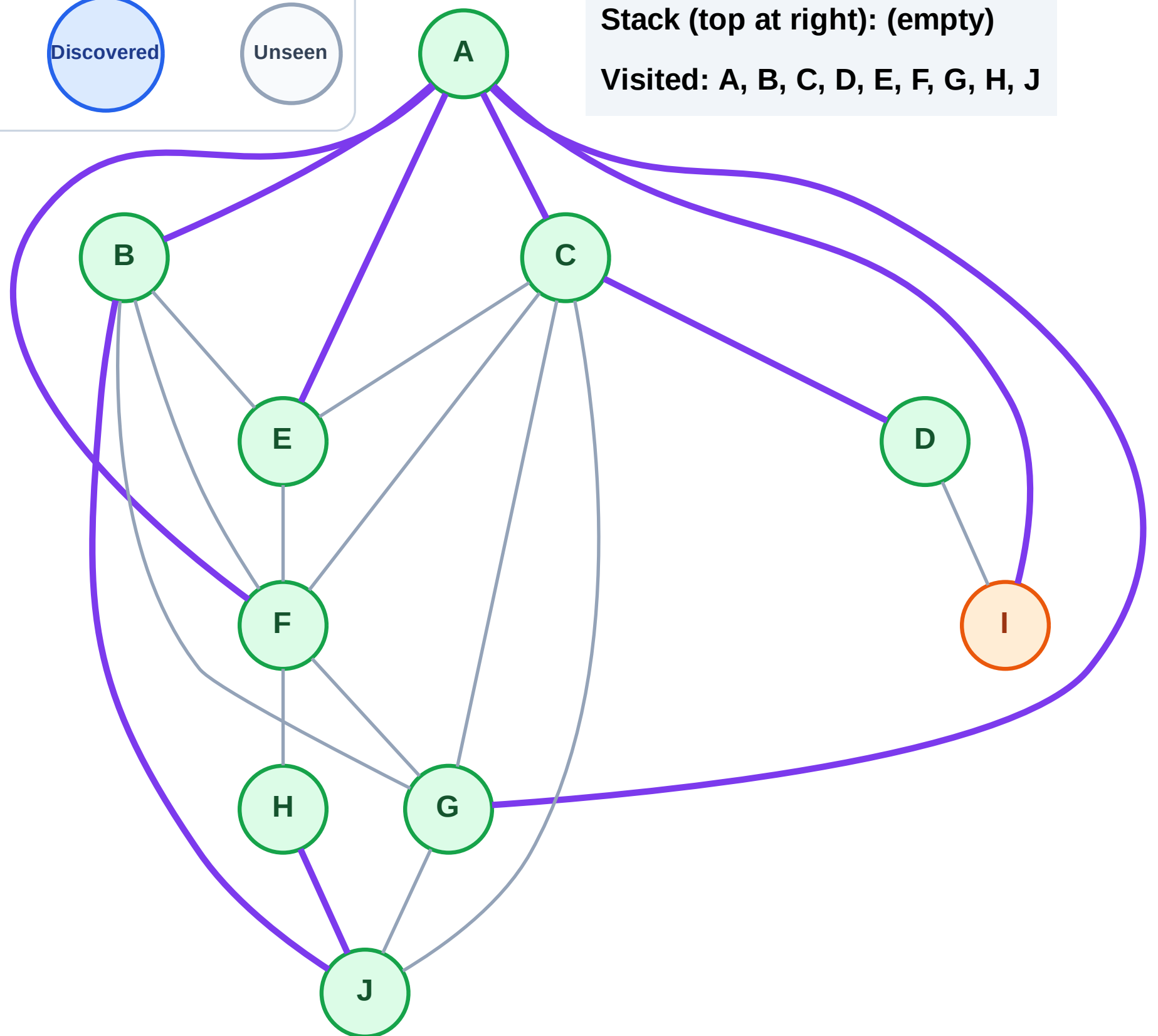
Processing

Discovered

Unseen

Stack (top at right): (empty)

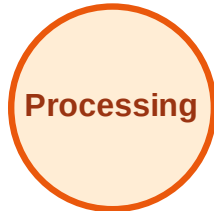
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

